WITTEN–PARKER–TAUBES PROOF OF THE POSITIVE ENERGY THEOREM

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Abstract. We give a self-contained exposition of the spinorial proof of the positive energy theorem, following Witten's argument together with the analytic completion of Parker and Taubes. Starting from a complete oriented asymptotically flat spacelike hypersurface (M, g, h) in a 4-dimensional Lorentz manifold satisfying the Einstein equation and the dominant energy condition, we define the hypersurface Dirac operator and derive its Weitzenböck formula. The integrated identity associated to this operator has a nonnegative bulk term and a boundary contribution at spatial infinity, whose leading part is the ADM energy-momentum. The analytic input is the solvability of the Witten equation in weighted Sobolev spaces with prescribed asymptotic constant spinors. Substituting these harmonic spinors into the integrated identity yields

$$E_\ell \geq |P_\ell|$$

on every asymptotically flat end M_ℓ.

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1. Asymptotically flat initial data and ADM energy-momentum

Throughout this article we work in geometrized units, so the gravitational constant is normalized to $G = 1$. Let $(N, g_{\mu\nu})$ be a 4-dimensional Lorentz manifold of signature $(-+++)$, and let $M \subset N$ be a complete oriented spacelike hypersurface. We denote by g the induced Riemannian metric on M, and by e_0 the future-pointing unit normal field along M.

If $\{e_1, e_2, e_3\}$ is a local g-orthonormal frame on TM, the second fundamental form of the embedding $M \hookrightarrow N$ is the symmetric tensor

$$h_{ij} = g\big(\nabla^N_{e_i} e_0, e_j\big)\,,$$

where ∇^N is the Levi-Civita connection of the spacetime metric. Thus an initial data set in our sense is the triple (M, g, h) arising from a spacelike hypersurface in a Lorentz manifold satisfying Einstein's equation

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi T_{\mu\nu}. \tag{1.1}$$

1.1. **Asymptotically flat ends.** The proof uses the geometry of M only through its asymptotic ends. We therefore begin by fixing the asymptotic hypotheses once and for all.

Definition 1.2. The initial data set (M, g, h) is called *asymptotically flat* if there exists a compact set $K \subset M$ such that

$$M \setminus K = M_1 \sqcup \cdots \sqcup M_k$$

is a finite disjoint union of connected subsets, called the *ends* of M, with the following property.

For each end M_ℓ there is a diffeomorphism from M_ℓ onto the complement of a contractible compact set in $\mathbb{R}^3$. In the corresponding Euclidean coordinates $x = (x^1, x^2, x^3)$, with $r = |x|$, the metric has the form

$$g_{ij} = \delta_{ij} + a_{ij}, \tag{1.3}$$

where

$$a_{ij} = O(r^{-1}), \qquad \partial_k a_{ij} = O(r^{-2}), \qquad \partial_\ell \partial_k a_{ij} = O(r^{-3}), \tag{1.4}$$

and the second fundamental form satisfies

$$h_{ij} = O(r^{-2}), \qquad \partial_k h_{ij} = O(r^{-3}). \tag{1.5}$$

The phrase "$O(r^{-m})$" is understood componentwise in the chosen asymptotic coordinates. Thus (1.4) means, for example, that there is a constant C such that

$$|a_{ij}(x)| \le Cr^{-1}, \qquad |\partial_k a_{ij}(x)| \le Cr^{-2}, \qquad |\partial_\ell \partial_k a_{ij}(x)| \le Cr^{-3}$$

for all x with r sufficiently large.

Since the compact set in the Euclidean model is contained in some ball $B_R = \{x \in \mathbb{R}^3 : |x| < R\}$, after enlarging R if necessary we may work on each end with the exterior region $\{r > R\}$. For $r > R$ we denote by

$$S_r = \{x \in \mathbb{R}^3 : |x| = r\}$$

the Euclidean coordinate sphere. We write ν for its outward Euclidean unit normal, dS for Euclidean surface measure, and

$$d\Omega^i := \nu^i \, dS. \tag{1.6}$$

Thus $d\Omega^i$ is the i-th Euclidean flux form on S_r.

The two decay assumptions (1.4) and (1.5) are tailored to the later boundary computation. The first derivative of the metric enters the energy integral, and the second fundamental form enters the momentum integral, so one needs exactly enough decay for those fluxes to have finite limits and for the error terms arising from the spinorial integration by parts to vanish at infinity.

Lemma 1.7. *On each asymptotically flat end,*

$$\big(\partial_j g_{ij} - \partial_i g_{jj}\big)\nu^i = O(r^{-2}), \tag{1.8}$$

and

$$\big(h_{ik} - \delta_{ik} h_{jj}\big)\nu^i = O(r^{-2}) \qquad \textit{for each } k = 1, 2, 3. \tag{1.9}$$

Proof. From (1.3) we have $g_{ij} = \delta_{ij} + a_{ij}$, hence

$$\partial_j g_{ij} - \partial_i g_{jj} = \partial_j a_{ij} - \partial_i a_{jj}.$$

By (1.4), each first derivative of a_{ij} is $O(r^{-2})$. Since $|\nu| = 1$, this gives (1.8).

For the second statement, (1.5) gives $h_{ik} = O(r^{-2})$. Taking the Euclidean trace,

$$h_{jj} = \delta^{ij} h_{ij},$$

we also have $h_{jj} = O(r^{-2})$. Therefore

$$h_{ik} - \delta_{ik} h_{jj} = O(r^{-2}),$$

and multiplying by the bounded vector ν^i yields (1.9). □

Because $\mathrm{Area}(S_r) \sim 4\pi r^2$, Lemma 1.7 shows that the corresponding flux integrals are of the correct order to admit finite limits. In the theorem below these limits are the total energy and total momentum of the end.

1.2. **Local energy density and the dominant energy condition.** The matter content of the spacetime is encoded in the symmetric tensor $T_{\mu\nu}$ in Einstein's equation (1.1). At a point $p \in M$, choose an adapted orthonormal frame

$$\{e_\alpha\}_{\alpha=0}^3 = \{e_0, e_1, e_2, e_3\},$$

where e_0 is the future-pointing unit normal to M and e_1, e_2, e_3 are tangent to M. We write

$$\rho := T(e_0, e_0), \qquad J_i := T(e_0, e_i). \tag{1.10}$$

Thus ρ is the local energy density seen by the normal observer e_0, and $J = (J_1, J_2, J_3)$ is the corresponding local momentum density along M.

The positivity hypothesis used in the proof is the dominant energy condition. Since this condition is most naturally stated in invariant language, we record that form first and then its expression in an adapted orthonormal frame.

Definition 1.11. We say that the *dominant energy condition* holds along M if for every point $p \in M$ and every future timelike vector $u \in T_pN$,

$$T(u, u) \geq 0,$$

and the covector $T(u, \cdot)$ is non-spacelike. Equivalently, for each adapted orthonormal frame $\{e_0, e_1, e_2, e_3\}$ along M,

$$T^{00} \geq |T^{\alpha\beta}| \qquad \text{for all } 0 \leq \alpha, \beta \leq 3, \tag{1.12}$$

and in particular

$$\rho \geq |J|_g. \tag{1.13}$$

Let us spell out why (1.13) follows from the geometric statement. In an adapted orthonormal frame, the covector $T(e_0, \cdot)$ has components

$$T(e_0, \cdot) = T_{00}\, e^0 + T_{0i}\, e^i.$$

Because the frame is orthonormal and the spacetime signature is $(- + ++)$, the Lorentzian square of this covector is

$$g^{-1}\big(T(e_0, \cdot), T(e_0, \cdot)\big) = -(T_{00})^2 + \sum_{i=1}^{3} (T_{0i})^2.$$

If $T(e_0, \cdot)$ is non-spacelike, this quantity is ≤ 0. Therefore

$$(T_{00})^2 \geq \sum_{i=1}^{3} (T_{0i})^2.$$

Since $T(e_0, e_0) = T_{00} = \rho \geq 0$, taking square roots gives

$$\rho \geq \left(\sum_{i=1}^{3} J_i^2 \right)^{1/2} = |J|_g.$$

This is exactly (1.13).

Later, when the curvature term in the Weitzenböck formula is rewritten using Einstein's equation, it becomes a Hermitian endomorphism involving T_{00} and T_{0i}. The nonnegativity of that endomorphism is precisely where the dominant energy condition enters the proof.

1.3. **ADM energy and momentum.** We now define the total energy and total momentum on each asymptotic end. These quantities depend only on the behavior of g and h near infinity.

Definition 1.14. Let M_ℓ be an asymptotically flat end. Its *ADM energy* is

$$E_\ell = \frac{1}{16\pi} \lim_{r \to \infty} \int_{S_r} \big(\partial_j g_{ij} - \partial_i g_{jj}\big)\, \nu^i \, dS. \tag{1.15}$$

Its *ADM momentum* is the vector $P_\ell = ((P_\ell)_1, (P_\ell)_2, (P_\ell)_3) \in \mathbb{R}^3$ with components

$$(P_\ell)_k = \frac{1}{8\pi} \lim_{r\to\infty} \int_{S_r} \big(h_{ik} - \delta_{ik} h_{jj}\big)\, \nu^i \, dS, \qquad k = 1, 2, 3. \tag{1.16}$$

Equivalently, using the flux forms (1.6),

$$E_\ell = \frac{1}{16\pi} \lim_{r\to\infty} \int_{S_r} \big(\partial_j g_{ij} - \partial_i g_{jj}\big)\, d\Omega^i, \tag{1.17}$$

and

$$(P_\ell)_k = \frac{1}{8\pi} \lim_{r\to\infty} \int_{S_r} \big(h_{ik} - \delta_{ik} h_{jj}\big)\, d\Omega^i. \tag{1.18}$$

The momentum formula (1.16) is written using the Euclidean tensor δ_{ij} associated with the chosen asymptotic coordinates. This is the form that enters most directly into the later boundary calculation. Under the decay assumptions (1.4)–(1.5), replacing δ_{ij} by g_{ij} in the trace term changes the integrand by $O(r^{-3})$, so the limiting value is unchanged. The point of the Euclidean form is that it is expressed solely in terms of the initial data (g, h) on M.

For later use we write

$$|P_\ell|^2 = \sum_{k=1}^{3} (P_\ell)_k^2. \tag{1.19}$$

1.4. **Statement of the theorem.** The quantity controlled by the spinor argument is the Lorentzian length of the energy-momentum vector (E_ℓ, P_ℓ). In the present normalization, the theorem takes the following form.

Theorem 1.20 (Positive energy theorem)**.** *Let (M, g, h) be a complete oriented asymptotically flat initial data set arising from a spacelike hypersurface in a 4-dimensional Lorentz manifold satisfying Einstein's equation (1.1). Assume the dominant energy condition from Definition 1.11. Then for each end M_ℓ,*

$$E_\ell \geq |P_\ell|. \tag{1.21}$$

The rest of the article is devoted to the proof of Theorem 1.20. The geometric part of the argument produces a Weitzenböck formula for a Dirac operator along the hypersurface M, together with an integral identity whose boundary term is exactly the flux appearing in Definition 1.14. The analytic part constructs harmonic spinors with prescribed constant limits on the asymptotic ends. Once those two ingredients are in place, the inequality (1.21) follows by choosing the asymptotic spinor appropriately on each end.

2. Dirac spinors along the hypersurface

2.1. **A linear algebra model for Dirac spinors.** Let $V = \mathbb{C}^2$ be the fundamental representation of $SL(2, \mathbb{C})$. This representation carries an invariant complex symplectic form, which we denote by

$$a : V \times V \longrightarrow \mathbb{C}.$$

Since a is nondegenerate, it identifies V with V^* up to the usual choice of index raising and lowering, but we will keep V and V^* notationally distinct.

A convenient concrete model for Minkowski space is the space of Hermitian 2×2 complex matrices. Given $x = (x^0, x^1, x^2, x^3) \in \mathbb{R}^4$, set

$$X(x) = \begin{pmatrix} x^0 + x^1 & x^2 + ix^3 \\ x^2 - ix^3 & x^0 - x^1 \end{pmatrix}. \tag{2.1}$$

The matrix $X(x)$ is Hermitian, and

$$\det X(x) = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2. \tag{2.2}$$

Thus, with our spacetime signature $(-+++)$,

$$\det X(x) = -\,g(x,x). \tag{2.3}$$

In this way we regard $\mathbb{R}^{3,1}$ as a real 4-dimensional subspace of $\mathrm{Hom}_{\mathbb{C}}(V^*, V)$.

The Dirac spinor representation is

$$S = V \oplus V^*. \tag{2.4}$$

If $x \in \mathbb{R}^{3,1} \subset \mathrm{Hom}_{\mathbb{C}}(V^*, V)$, let

$$x' : V \longrightarrow V^*$$

denote the a-adjoint of x, characterized by

$$a(x\eta, \xi) = a(\eta, x'\xi) \qquad \text{for all } \eta \in V^*,\ \xi \in V. \tag{2.5}$$

Clifford multiplication by x on S is then defined by

$$x \cdot (\xi, \eta) = \big(x\eta,\ x'\xi\big), \qquad (\xi, \eta) \in V \oplus V^*. \tag{2.6}$$

We now check the Clifford relations. Applying $x\cdot$ twice gives

$$x \cdot x \cdot (\xi, \eta) = \big(xx'\xi,\ x'x\eta\big).$$

Because V and V^* are 2-dimensional, the endomorphisms xx' and $x'x$ are scalar multiples of the identity once one knows their determinants. In the matrix model (2.1), a direct calculation gives

$$xx' = x'x = \det X(x)\,\mathrm{Id}.$$

Using (2.3), we obtain

$$x \cdot x\cdot = -\,g(x,x)\,\mathrm{Id}_S.$$

Polarizing this identity yields the Clifford relations

$$x \cdot y \cdot + y \cdot x\cdot = -2\,g(x,y)\,\mathrm{Id}_S, \qquad x, y \in \mathbb{R}^{3,1}. \tag{2.7}$$

2.2. **The two Hermitian forms.** There is an $SL(2,\mathbb{C})$-invariant Hermitian form $(\,\cdot\,,\,\cdot\,)$ on S. We do not need its explicit coordinate formula; what matters in the sequel is that it is nondegenerate, indefinite, and satisfies

$$(x \cdot \psi, \phi) = (\psi, x \cdot \phi) \qquad \text{for all } x \in \mathbb{R}^{3,1},\ \psi, \phi \in S. \tag{2.8}$$

Thus Clifford multiplication is Hermitian with respect to $(\,\cdot\,,\,\cdot\,)$.

To obtain a positive definite inner product, fix a unit future timelike covector e^0. Since $g(e^0, e^0) = -1$, the Clifford relations give

$$(e^0\cdot)^2 = \mathrm{Id}_S. \tag{2.9}$$

Definition 2.10. For $\psi, \phi \in S$, define

$$\langle \psi, \phi \rangle := (e^0 \cdot \psi, \phi). \tag{2.11}$$

Lemma 2.12. *The Hermitian form $\langle\,\cdot\,,\,\cdot\,\rangle$ has the following properties.*

(1) It is positive definite.
(2) The operator $e^0\cdot$ is Hermitian with respect to $\langle\,\cdot\,,\,\cdot\,\rangle$.
(3) If $x \in \mathbb{R}^{3,1}$ satisfies $g(x, e^0) = 0$, then $x\cdot$ is skew-Hermitian with respect to $\langle\,\cdot\,,\,\cdot\,\rangle$:

$$\langle x \cdot \psi, \phi\rangle = -\langle \psi, x \cdot \phi\rangle.$$

(4) If $x, y \in \mathbb{R}^{3,1}$ satisfy $g(x, e^0) = g(y, e^0) = 0$, then the Clifford commutator

$$[x, y] := x \cdot y \cdot - y \cdot x\cdot$$

is Hermitian with respect to $\langle\,\cdot\,,\,\cdot\,\rangle$.

Proof. We begin with positivity. The timelike covector e^0 identifies V^* with V by Clifford multiplication,

$$e^0\cdot : V^* \longrightarrow V.$$

Choose a basis in which this map is represented by the identity. In that basis, a spinor $\psi \in S = V \oplus V^*$ has the form $\psi = (\xi, \eta)$, and the definition (2.11) becomes

$$\langle \psi, \psi\rangle = |\xi|^2 + |\eta|^2.$$

In particular $\langle \psi, \psi\rangle > 0$ whenever $\psi \neq 0$, so $\langle\,\cdot\,,\,\cdot\,\rangle$ is positive definite.

Next, using (2.8) and (2.9),

$$\langle e^0 \cdot \psi, \phi\rangle = (e^0 \cdot e^0 \cdot \psi, \phi) = (\psi, \phi) = (e^0 \cdot \psi, e^0 \cdot \phi) = \langle \psi, e^0 \cdot \phi\rangle.$$

Thus $e^0\cdot$ is Hermitian.

Now let x be orthogonal to e^0. Then the Clifford relations give

$$e^0 \cdot x \cdot + x \cdot e^0\cdot = -2\, g(e^0, x)\,\mathrm{Id} = 0,$$

so

$$e^0 \cdot x\cdot = -x \cdot e^0 \cdot . \tag{2.13}$$

Therefore

$$\langle x \cdot \psi, \phi\rangle = (e^0 \cdot x \cdot \psi, \phi) = (-x \cdot e^0 \cdot \psi, \phi) = -(e^0 \cdot \psi, x \cdot \phi) = -\langle \psi, x \cdot \phi\rangle,$$

which proves the skew-Hermitian statement.

Finally, if $x\cdot$ and $y\cdot$ are skew-Hermitian, then

$$(x \cdot y\cdot)^* = y \cdot x\cdot, \qquad (y \cdot x\cdot)^* = x \cdot y \cdot .$$

Hence

$$[x, y]^* = (x \cdot y \cdot - y \cdot x\cdot)^* = y \cdot x \cdot - x \cdot y\cdot = [x, y],$$

so $[x, y]$ is Hermitian. □

From now on, all L^2 norms and all formal adjoints will be taken with respect to the positive definite inner product $\langle\,\cdot\,,\,\cdot\,\rangle$.

2.3. **Spin structures on oriented 3-manifolds.** Let $F(M) \to M$ be the principal $SO(3)$-bundle of oriented orthonormal frames of TM. A spin structure on M is a principal Spin(3)-bundle

$$P(M) \longrightarrow M$$

together with a $\mathrm{Spin}(3) \to SO(3)$-equivariant double covering

$$P(M) \longrightarrow F(M).$$

We use the following standard fact.

Proposition 2.14. *Every oriented 3-manifold admits a spin structure. Equivalently, if M is oriented, then $w_2(M) = 0$.*

Fix such a spin structure once and for all. Since M is a spacelike hypersurface in N, an oriented orthonormal frame of TM together with the future-pointing unit normal e_0 determines an orthonormal Lorentz frame of $TN|_M$. On the level of principal bundles, this identifies the pullback of the Lorentz frame bundle to M with the extension of structure group of $F(M)$ from $SO(3)$ to $SO(3,1)$. Lifting this picture to the double covers gives a principal $\mathrm{Spin}(3,1)$-bundle over M, and hence the associated vector bundle

$$S = P(M) \times_\rho (V \oplus V^*), \tag{2.15}$$

where ρ is the Dirac spinor representation of $\mathrm{Spin}(3,1) \cong SL(2,\mathbb{C})$. By abuse of notation we denote the representation and the resulting vector bundle by the same letter S.

Sections of S will be called *Dirac spinors along M*. Fiberwise Clifford multiplication extends from $\mathbb{R}^{3,1}$ to $T^*N|_M$, so if $\alpha \in T^*N|_M$ and $\psi \in \Gamma(S)$, the product $\alpha \cdot \psi$ is again a section of S. The two Hermitian forms $(\,\cdot\,,\,\cdot\,)$ and $\langle\,\cdot\,,\,\cdot\,\rangle$ also extend fiberwise to the bundle S.

2.4. **Constant spinors on the asymptotic ends.** Let M_ℓ be one of the asymptotically flat ends from Definition 1.2, and let

$$\Phi_\ell : \mathbb{R}^3 \setminus K_\ell \longrightarrow M_\ell$$

be the diffeomorphism defining that end, where $K_\ell \subset \mathbb{R}^3$ is contractible and compact.

The pullback spin bundle $\Phi_\ell^* P(M)$ is a spin structure on $\mathbb{R}^3 \setminus K_\ell$. Spin structures on a connected manifold are classified by $H^1(\,\cdot\,;\mathbb{Z}_2)$, and for the present end this cohomology group vanishes:

$$H^1(\mathbb{R}^3 \setminus K_\ell; \mathbb{Z}_2) = 0.$$

Consequently $\Phi_\ell^* P(M)$ is isomorphic to the trivial spin bundle on $\mathbb{R}^3 \setminus K_\ell$.

The associated spinor bundle $\Phi_\ell^* S$ is therefore trivial over the end and extends trivially across K_ℓ to the product bundle

$$\mathbb{R}^3 \times (V \oplus V^*).$$

A section of this product bundle is called *constant* if its value in $V \oplus V^*$ is independent of the point of $\mathbb{R}^3$. Pulling such sections back by Φ_ℓ^{-1} gives a distinguished family of spinors on M_ℓ.

Definition 2.16. A *constant spinor on the end* M_ℓ is a section of $S|_{M_\ell}$ obtained by pulling back a constant section of the trivial bundle

$$\mathbb{R}^3 \times (V \oplus V^*) \longrightarrow \mathbb{R}^3$$

under the chosen asymptotic identification Φ_ℓ.

This notion is the boundary condition used later when solving the hypersurface Dirac equation. The point is that the spinor itself is not required to be covariantly constant with respect to the curved connection on M_ℓ; it is only required to become constant in the asymptotic trivialization determined by the end.

2.5. **The two spin connections.** The spinor bundle $S \to M$ carries two natural connections.

The first comes from the Lorentzian Levi-Civita connection of N. Let ω_{ab} be the connection 1-forms of the spacetime Levi-Civita connection in a local orthonormal frame $\{e_a\}_{a=0}^3$ adapted to M, with e_0 normal and e_1, e_2, e_3 tangent. The induced connection on spinors is

$$V_X\psi = X(\psi) - \frac{1}{4}\,\omega_{ab}(X)\,e^a \cdot e^b \cdot \psi, \qquad X \in TM. \tag{2.17}$$

Because it is induced from the Spin$(3,1)$-structure, V preserves the indefinite form $(\,\cdot\,,\,\cdot\,)$.

The second connection is induced from the Riemannian Levi-Civita connection of (M, g). If θ_{ij} are the connection 1-forms of M in a local oriented orthonormal frame $\{e_i\}_{i=1}^3$, then

$$\nabla_X\psi = X(\psi) - \frac{1}{4}\,\theta_{ij}(X)\,e^i \cdot e^j \cdot \psi, \qquad X \in TM. \tag{2.18}$$

Since ∇ is induced from the Spin(3)-structure determined by g, it preserves the positive definite inner product $\langle\,\cdot\,,\,\cdot\,\rangle$.

Thus V and ∇ are not the same connection, and they preserve different fiber metrics. The relation between them is exactly the place where the second fundamental form enters the spinorial argument.

Lemma 2.19. *Let $\{e_0, e_1, e_2, e_3\}$ be an adapted local orthonormal frame along M, and let*

$$h_{ij} = g(\nabla^N_{e_i} e_0, e_j)$$

be the second fundamental form of $M \subset N$. Then for every $\psi \in \Gamma(S)$,

$$V_i\psi = \nabla_i\psi + \frac{1}{2}\,h_{ij}\,e^j \cdot e^0 \cdot \psi, \qquad V_i := V_{e_i},\ \nabla_i := \nabla_{e_i}. \tag{2.20}$$

Proof. We compare (2.17) and (2.18) for $X = e_i$.

First, because the tangential part of the spacetime Levi-Civita connection coincides with the Levi-Civita connection of M, the tangential connection forms agree on TM:

$$\omega_{jk}(e_i) = \theta_{jk}(e_i).$$

Second, the normal-tangential connection coefficients are determined by the second fundamental form. Indeed,

$$\nabla^N_{e_i} e_0 = h_{ij} e_j,$$

so by definition of the connection 1-forms,

$$\omega_{0j}(e_i) = g(\nabla^N_{e_i} e_0, e_j) = h_{ij}.$$

Using metric compatibility and the skew-symmetry $\omega_{ab} = -\omega_{ba}$, we also have

$$\omega_{j0}(e_i) = -\,\omega_{0j}(e_i) = -h_{ij}.$$

Substituting these identities into (2.17) gives

$$\begin{aligned} V_i\psi &= e_i(\psi) - \frac{1}{4}\,\omega_{jk}(e_i)\,e^j\cdot e^k\cdot\psi - \frac{1}{4}\,\omega_{0j}(e_i)\,e^0\cdot e^j\cdot\psi - \frac{1}{4}\,\omega_{j0}(e_i)\,e^j\cdot e^0\cdot\psi \\ &= e_i(\psi) - \frac{1}{4}\,\theta_{jk}(e_i)\,e^j\cdot e^k\cdot\psi - \frac{1}{4}\,h_{ij}\,e^0\cdot e^j\cdot\psi + \frac{1}{4}\,h_{ij}\,e^j\cdot e^0\cdot\psi. \end{aligned}$$

The first two terms are exactly $\nabla_i\psi$, so

$$V_i\psi = \nabla_i\psi + \frac{1}{4}\,h_{ij}\big(e^j\cdot e^0\cdot - e^0\cdot e^j\cdot\big)\psi.$$

Because $g(e^0, e^j) = 0$, the Clifford relations give

$$e^0\cdot e^j\cdot + e^j\cdot e^0\cdot = 0,$$

hence

$$e^j\cdot e^0\cdot - e^0\cdot e^j\cdot = 2\,e^j\cdot e^0\cdot\,.$$

Therefore

$$V_i\psi = \nabla_i\psi + \frac{1}{2}\,h_{ij}\,e^j\cdot e^0\cdot\psi,$$

which is (2.20). □

The formula (2.20) explains why the later Dirac operator is not the intrinsic Dirac operator of (M, g). Its principal symbol will be the same, since both V and ∇ differentiate only once and use the same Clifford multiplication by tangent covectors. The difference lies in the lower-order term involving the second fundamental form h_{ij}.

2.6. **The operator used later.** The spinorial proof does not use the Dirac operator formed from ∇. Instead it uses the operator obtained by Clifford multiplying the ambient-induced connection V by the tangential coframe:

$$\sum_{i=1}^{3} e^i\cdot V_i.$$

Its symbol is the same as that of the intrinsic Dirac operator

$$\sum_{i=1}^{3} e^i\cdot\nabla_i,$$

but the two operators differ by the contribution of the second fundamental form encoded in (2.20). That operator will be introduced and analyzed in the next section.

3. The hypersurface Dirac operator and the Weitzenböck formula

Let $S \to M$ be the Dirac spinor bundle from Section 2. It carries the ambient-induced connection V and the intrinsic Riemannian spin connection ∇, related by

$$V_i\psi = \nabla_i\psi + \frac{1}{2}\,h_{ij}\,e^j\cdot e^0\cdot\psi. \tag{3.1}$$

Throughout this section $\langle\,\cdot\,,\,\cdot\,\rangle$ denotes the positive definite inner product from Definition 2.10, and μ denotes the Riemannian volume form of (M, g).

3.1. **The hypersurface Dirac operator.**

Definition 3.2. The *hypersurface Dirac operator* is the first-order differential operator

$$\mathscr{D}\psi = \sum_{i=1}^{3} e^i \cdot V_i \psi. \tag{3.3}$$

Its principal symbol is the same as that of the intrinsic Dirac operator of (M, g), since only the zeroth-order part of the connection changes when one replaces ∇ by V. Using (3.1), one can write $\mathscr{D}$ explicitly in terms of the intrinsic Dirac operator.

Lemma 3.4. *Let*

$$D_M \psi := \sum_{i=1}^{3} e^i \cdot \nabla_i \psi$$

be the intrinsic Dirac operator of the Riemannian manifold (M, g)*. Then*

$$\mathscr{D} = D_M - \frac{1}{2}(\mathrm{tr}_g h)\, e^0 \cdot . \tag{3.5}$$

Proof. Substituting (3.1) into (3.3) gives

$$\mathscr{D}\psi = \sum_{i=1}^{3} e^i \cdot \nabla_i \psi + \frac{1}{2} \sum_{i,j=1}^{3} h_{ij}\, e^i \cdot e^j \cdot e^0 \cdot \psi.$$

The first term is $D_M \psi$. To simplify the second term, split the double sum into diagonal and off-diagonal parts:

$$\sum_{i,j} h_{ij}\, e^i \cdot e^j \cdot = \sum_i h_{ii}\, e^i \cdot e^i \cdot + \sum_{i<j} h_{ij} \big(e^i \cdot e^j \cdot + e^j \cdot e^i \cdot \big).$$

Since $h_{ij} = h_{ji}$ and the tangential Clifford relations give

$$e^i \cdot e^j \cdot + e^j \cdot e^i \cdot = -2\delta^{ij} \mathrm{Id},$$

the off-diagonal sum vanishes, while

$$e^i \cdot e^i \cdot = -\mathrm{Id}.$$

Hence

$$\sum_{i,j} h_{ij}\, e^i \cdot e^j \cdot = -\sum_i h_{ii}\, \mathrm{Id} = -(\mathrm{tr}_g h)\, \mathrm{Id},$$

and therefore

$$\mathscr{D}\psi = D_M \psi - \frac{1}{2}(\mathrm{tr}_g h)\, e^0 \cdot \psi.$$

□

Since the intrinsic Dirac operator D_M is formally self-adjoint with respect to $\langle \cdot, \cdot \rangle$, and $e^0 \cdot$ is Hermitian by Lemma 2.12, the operator $\mathscr{D}$ is also formally self-adjoint. We record the resulting Green formula.

Lemma 3.6. *Let* $\Omega \subset M$ *be a compact domain with smooth boundary, and let* ν *be the outward unit normal to* $\partial\Omega$*. Then for all* $\phi, \psi \in \Gamma(S)$*,*

$$\int_\Omega \big\langle \mathscr{D}\phi, \psi \big\rangle\, \mu - \int_\Omega \big\langle \phi, \mathscr{D}\psi \big\rangle\, \mu = \int_{\partial\Omega} \big\langle \nu \cdot \phi, \psi \big\rangle\, d\sigma. \tag{3.7}$$

In particular, $\mathscr{D}^* = \mathscr{D}$*.*

Proof. By (3.5),

$$\mathscr{D} = D_M - \frac{1}{2}(\mathrm{tr}_g h)e^0 \cdot .$$

The intrinsic Dirac operator satisfies the usual Green formula

$$\int_\Omega \langle D_M\phi, \psi\rangle\, \mu - \int_\Omega \langle \phi, D_M\psi\rangle\, \mu = \int_{\partial\Omega} \langle \nu\cdot\phi, \psi\rangle\, d\sigma.$$

The zeroth-order term contributes

$$-\frac{1}{2}\int_\Omega (\mathrm{tr}_g h)\, \langle e^0\cdot\phi, \psi\rangle\, \mu + \frac{1}{2}\int_\Omega (\mathrm{tr}_g h)\, \langle \phi, e^0\cdot\psi\rangle\, \mu,$$

and these two terms cancel because $e^0\cdot$ is Hermitian. This gives (3.7). □

3.2. **An adapted moving frame.** We now fix the local frame used in the computation of $\mathscr{D}^2$. Let $p \in M$. Choose an orthonormal basis $\{e_0, e_1, e_2, e_3\}$ of T_pN with e_0 normal to M and $e_1, e_2, e_3 \in T_pM$. Extend e_1, e_2, e_3 to a local orthonormal frame on M so that

$$(\nabla_{e_i} e_j)_p = 0 \qquad \text{for } 1 \le i, j \le 3. \tag{3.8}$$

Extend further to a local orthonormal frame $\{e_0, e_1, e_2, e_3\}$ of TN near p with e_0 normal to M along M. Let $\{e^0, e^1, e^2, e^3\}$ be the dual coframe.

At the point p, the frame has three useful properties. First,

$$[e_i, e_j]_p = 0, \qquad 1 \le i, j \le 3, \tag{3.9}$$

because (3.8) says the Levi-Civita connection of M vanishes there in this frame. Second, since

$$\nabla^N_{e_i} e_0 = h_{ij} e_j,$$

metric compatibility gives

$$\nabla^N_{e_i} e_j = h_{ij} e_0 \qquad \text{at } p,$$

because the tangential part vanishes by (3.8). Third, differentiating the dual coframe with the ambient connection gives the identities below.

Lemma 3.10. *At the point p,*

$$(V_i e^0)_p = -\, h_{ij}\, e^j, \tag{3.11}$$

and

$$(V_i e^j)_p = -\, h_{ij}\, e^0. \tag{3.12}$$

Proof. Recall that V acts on covectors by the ambient Levi-Civita connection. Thus for any vector field X,

$$(V_i e^0)(X) = e_i\big(e^0(X)\big) - e^0\big(\nabla^N_{e_i} X\big).$$

Evaluate first on $X = e_k$. Since $e^0(e_k) = 0$,

$$(V_i e^0)(e_k) = -\, e^0(\nabla^N_{e_i} e_k) = -\, e^0(h_{ik} e_0) = -\, h_{ik}.$$

Evaluating on $X = e_0$ gives

$$(V_i e^0)(e_0) = e_i(e^0(e_0)) - e^0(\nabla^N_{e_i} e_0) = 0 - e^0(h_{ij} e_j) = 0.$$

Hence $(V_i e^0)_p = -h_{ij} e^j$.

Similarly,

$$(V_i e^j)(X) = e_i\big(e^j(X)\big) - e^j\big(\nabla^N_{e_i} X\big).$$

Evaluating on $X = e_k$, we get

$$(V_i e^j)(e_k) = -\, e^j(\nabla^N_{e_i} e_k) = -\, e^j(h_{ik} e_0) = 0.$$

Evaluating on $X = e_0$, we obtain

$$(V_i e^j)(e_0) = -\, e^j(\nabla^N_{e_i} e_0) = -\, e^j(h_{ik} e_k) = -\, h_{ij}.$$

Therefore $(V_i e^j)_p = -h_{ij} e^0$. □

3.3. **Expanding $\mathscr{D}^2$.** We now compute $\mathscr{D}^2$ at the point p using the adapted frame above. Starting from (3.3),

$$\begin{aligned} \mathscr{D}^2\psi &= \sum_{i,j=1}^{3} e^i \cdot V_i\big(e^j \cdot V_j\psi\big) \\ &= \sum_{i,j} e^i \cdot (V_i e^j) \cdot V_j\psi + \sum_{i,j} e^i \cdot e^j \cdot V_i V_j \psi. \end{aligned} \tag{3.13}$$

Using (3.12), the first sum becomes

$$\sum_{i,j} e^i \cdot (V_i e^j) \cdot V_j\psi = -\sum_{i,j} h_{ij}\, e^i \cdot e^0 \cdot V_j\psi. \tag{3.14}$$

To treat the second sum in (3.13), separate the diagonal and off-diagonal parts:

$$\begin{aligned} \sum_{i,j} e^i \cdot e^j \cdot V_i V_j \psi &= \sum_i e^i \cdot e^i \cdot V_i V_i \psi + \sum_{i<j}\big(e^i \cdot e^j \cdot V_i V_j \psi + e^j \cdot e^i \cdot V_j V_i \psi\big) \\ &= -\sum_i V_i V_i \psi + \sum_{i<j} e^i \cdot e^j \cdot (V_i V_j - V_j V_i)\psi. \end{aligned} \tag{3.15}$$

At the point p, $[e_i, e_j]_p = 0$, so

$$(V_i V_j - V_j V_i)\psi = \Omega_{ij}\psi,$$

where Ω_{ij} is the curvature of the spin connection V. With our sign convention,

$$\Omega_{ij}\psi = -\frac{1}{4}\, R_{ij\alpha\beta}\, e^\alpha \cdot e^\beta \cdot \psi, \tag{3.16}$$

where $R_{\alpha\beta\gamma\delta}$ is the Riemann curvature tensor of the spacetime metric.

Substituting (3.14), (3.15), and (3.16) into (3.13) gives

$$\mathscr{D}^2\psi = -\sum_i V_i V_i \psi - \sum_{i,j} h_{ij}\, e^i \cdot e^0 \cdot V_j\psi - \frac{1}{4}\sum_{i<j} R_{ij\alpha\beta}\, e^i \cdot e^j \cdot e^\alpha \cdot e^\beta \cdot \psi. \tag{3.17}$$

The last term is the curvature contribution. It can be simplified using only the algebraic symmetries of the Riemann tensor and the first Bianchi identity. We record the resulting contraction separately.

Lemma 3.18. *At the point p,*

$$-\frac{1}{4}\sum_{i<j} R_{ij\alpha\beta}\, e^i \cdot e^j \cdot e^\alpha \cdot e^\beta = \frac{1}{4}\Big(R + 2R_{00} + 2R_{0j}\, e^0 \cdot e^j\Big), \tag{3.19}$$

where R is the spacetime scalar curvature and $R_{\mu\nu}$ is the spacetime Ricci tensor in the adapted orthonormal frame.

Proof. Set

$$C := -\frac{1}{4}\sum_{i<j} R_{ij\alpha\beta}\, e^i \cdot e^j \cdot e^\alpha \cdot e^\beta.$$

Because i, j are tangential and because the Riemann tensor is skew in each index pair, every term in C is built from products of either four tangential Clifford factors or two tangential factors together with one factor e^0. The first Bianchi identity removes the degree-four tangential part, so only a scalar term and a mixed term proportional to $e^0 \cdot e^j$ remain. Accordingly,

$$C = A\,\mathrm{Id} + B_j\, e^0 \cdot e^j$$

for some coefficients A and B_j.

To determine the scalar part, one keeps only those terms in which the Clifford product reduces to a multiple of the identity. Using

$$e^i \cdot e^j \cdot e^i \cdot e^j = -\mathrm{Id} \qquad (i \neq j),$$

and summing over the relevant curvature components gives

$$A = \frac{1}{4}(R + 2R_{00}).$$

To determine the mixed term, one collects the contributions containing exactly one factor e^0 after Clifford simplification. Using the skew-symmetry of the curvature tensor and the Bianchi identity, these terms combine to

$$B_j = \frac{1}{2}R_{0j}.$$

Substituting these values of A and B_j proves (3.19). □

Combining (3.17) and (3.19), we obtain the local expansion of $\mathscr{D}^2$.

Proposition 3.20. *At every point of M,*

$$\mathscr{D}^2\psi = -\sum_{i=1}^{3} V_i V_i \psi - \sum_{i,j=1}^{3} h_{ij}\, e^i \cdot e^0 \cdot V_j \psi + \frac{1}{4}\Big(R + 2R_{00} + 2R_{0j}\, e^0 \cdot e^j\Big)\psi. \tag{3.21}$$

3.4. **Formal adjoints.** The next step is to identify the formal adjoint of V with respect to the positive definite inner product $\langle\,\cdot\,,\,\cdot\,\rangle$. Because V does not preserve $\langle\,\cdot\,,\,\cdot\,\rangle$, its adjoint is not simply $-V$.

Lemma 3.22. *Let $\phi, \psi \in \Gamma(S)$. In the adapted frame above, one has*

$$d\Big[\langle \phi, V_i\psi\rangle\, e_i \lrcorner \mu\Big] = \Big(\langle V_i\phi, V_i\psi\rangle + \big\langle \phi,\, V_iV_i\psi + h_{ij}\, e^j \cdot e^0 \cdot V_i\psi\big\rangle\Big)\mu. \tag{3.23}$$

Consequently,

$$V_i^* = -\,V_i - h_{ij}\, e^j \cdot e^0 \cdot, \tag{3.24}$$

and therefore

$$V^*V = -\sum_i V_iV_i - \sum_{i,j} h_{ij}\, e^j \cdot e^0 \cdot V_i. \tag{3.25}$$

Proof. At the point p, because $(\nabla_{e_i} e_i)_p = 0$, we have

$$d\Big[\langle \phi, V_i\psi\rangle\, e_i \lrcorner \mu\Big] = e_i\big(\langle \phi, V_i\psi\rangle\big)\, \mu.$$

Now

$$\langle \phi, V_i\psi\rangle = (e^0 \cdot \phi, V_i\psi),$$

where $(\,\cdot\,,\,\cdot\,)$ is the indefinite Hermitian form preserved by V. Differentiating and using that V preserves $(\,\cdot\,,\,\cdot\,)$, we obtain

$$\begin{aligned} e_i\big(\langle \phi, V_i\psi\rangle\big) &= \big((V_i e^0)\cdot\phi, V_i\psi\big) + \big(e^0\cdot V_i\phi, V_i\psi\big) + \big(e^0\cdot\phi, V_iV_i\psi\big) \\ &= \big((V_i e^0)\cdot\phi, V_i\psi\big) + \langle V_i\phi, V_i\psi\rangle + \langle \phi, V_iV_i\psi\rangle. \end{aligned}$$

Using (3.11),

$$(V_i e^0)\cdot\phi = -h_{ij}\, e^j\cdot\phi,$$

hence

$$\big((V_i e^0)\cdot\phi, V_i\psi\big) = -h_{ij}\,(e^j\cdot\phi, V_i\psi).$$

Because Clifford multiplication is Hermitian with respect to $(\,\cdot\,,\,\cdot\,)$,

$$(e^j\cdot\phi, V_i\psi) = (\phi, e^j\cdot V_i\psi).$$

Also, for any spinors η, χ,

$$(\eta, \chi) = \langle \eta, e^0\cdot\chi\rangle.$$

Therefore

$$-h_{ij}(e^j\cdot\phi, V_i\psi) = -h_{ij}\,\langle \phi, e^0\cdot e^j\cdot V_i\psi\rangle = h_{ij}\,\langle \phi, e^j\cdot e^0\cdot V_i\psi\rangle,$$

since $e^0\cdot e^j = -e^j\cdot e^0$ for tangential e^j. Substituting this back proves (3.23).

Now integrate (3.23) over a compact domain and compare with the definition of the formal adjoint. This gives

$$V_i^* = -V_i - h_{ij}\, e^j\cdot e^0\cdot\,.$$

Summing over i yields (3.25). □

3.5. **The Weitzenböck formula.** Comparing (3.21) and (3.25), and using the symmetry $h_{ij} = h_{ji}$, we find that the second-order and first-order terms agree exactly. This leaves only the curvature endomorphism.

Theorem 3.26 (Weitzenböck formula)**.** *The hypersurface Dirac operator satisfies*

$$\mathscr{D}^2 = V^*V + \mathcal{R}, \tag{3.27}$$

where

$$\mathcal{R} = \frac{1}{4}\Big(R + 2R_{00} + 2R_{0j}\, e^0\cdot e^j\Big) \in \mathrm{End}(S). \tag{3.28}$$

Since $\mathscr{D}$ is formally self-adjoint, this may also be written as

$$\mathscr{D}^*\mathscr{D} = V^*V + \mathcal{R}. \tag{3.29}$$

Proof. By Proposition 3.20,

$$\mathscr{D}^2 = -\sum_i V_iV_i - \sum_{i,j} h_{ij}\, e^i\cdot e^0\cdot V_j + \mathcal{R}.$$

By (3.25),

$$V^*V = -\sum_i V_iV_i - \sum_{i,j} h_{ij}\, e^j\cdot e^0\cdot V_i.$$

Because $h_{ij} = h_{ji}$, renaming the dummy indices $i \leftrightarrow j$ in the second expression shows

$$\sum_{i,j} h_{ij}\, e^j \cdot e^0 \cdot V_i = \sum_{i,j} h_{ij}\, e^i \cdot e^0 \cdot V_j.$$

Thus the first-order terms coincide, and (3.27) follows. The identity (3.29) is equivalent because $\mathscr{D}^* = \mathscr{D}$. □

3.6. **The curvature term and the energy condition.** The endomorphism $\mathcal{R}$ becomes transparent after one substitutes Einstein's equation. Recall from Section 1 that

$$\rho = T(e_0, e_0), \qquad J_i = T(e_0, e_i).$$

Lemma 3.30. *Along M, Einstein's equation implies*

$$\mathcal{R} = 4\pi\Big(\rho + J_i\, e^0 \cdot e^i\Big). \tag{3.31}$$

Proof. From Einstein's equation

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi T_{\mu\nu},$$

we obtain, in the adapted orthonormal frame,

$$R_{00} + \frac{1}{2} R = 8\pi T_{00} = 8\pi\rho$$

because $g_{00} = -1$, and

$$R_{0i} = 8\pi T_{0i} = 8\pi J_i$$

because $g_{0i} = 0$. Substituting these into (3.28) gives

$$\mathcal{R} = \frac{1}{4}\Big(16\pi\rho + 16\pi J_i\, e^0 \cdot e^i\Big) = 4\pi\Big(\rho + J_i\, e^0 \cdot e^i\Big).$$

□

Lemma 3.32. *If the dominant energy condition holds, then $\mathcal{R}$ is pointwise nonnegative:*

$$\langle \psi, \mathcal{R}\psi \rangle \geq 0 \qquad \textit{for all } \psi \in S. \tag{3.33}$$

Proof. Set

$$B := J_i\, e^0 \cdot e^i.$$

Each $e^0 \cdot e^i$ is Hermitian with respect to $\langle \cdot\,, \cdot \rangle$ by Lemma 2.12, so B is Hermitian. We compute its square:

$$B^2 = J_i J_j\, e^0 \cdot e^i \cdot e^0 \cdot e^j.$$

Since $e^0 \cdot e^i = -e^i \cdot e^0$ and $(e^0\cdot)^2 = \mathrm{Id}$,

$$e^0 \cdot e^i \cdot e^0 \cdot e^j = -\, e^i \cdot e^j.$$

Hence

$$B^2 = -\, J_i J_j\, e^i \cdot e^j.$$

Now split the sum into diagonal and off-diagonal parts. The off-diagonal part vanishes because $J_i J_j$ is symmetric in i, j, while $e^i \cdot e^j + e^j \cdot e^i = 0$ for $i \neq j$. For the diagonal part,

$$-\, J_i^2\, e^i \cdot e^i = -\, J_i^2(-\mathrm{Id}) = J_i^2 \mathrm{Id}.$$

Therefore

$$B^2 = |J|_g^2\, \mathrm{Id}. \tag{3.34}$$

Since B is Hermitian, its eigenvalues are $\pm|J|_g$. Thus the eigenvalues of $\rho\,\mathrm{Id} + B$ are $\rho \pm |J|_g$. By the dominant energy condition $\rho \geq |J|_g$, these are all nonnegative. Using (3.31), we conclude that $\mathcal{R}$ is nonnegative. □

3.7. **The integral identity.** The operator identity (3.27) yields the basic integral formula used later. We first write down the two divergence identities from which the boundary term arises.

Lemma 3.35. *For $\phi, \psi \in \Gamma(S)$, one has*

$$d\Big[\langle \phi, V_i\psi\rangle\, e_i \lrcorner \mu\Big] = \Big(\langle V_i\phi, V_i\psi\rangle - \langle \phi, V^*V\psi\rangle\Big)\mu, \tag{3.36}$$

and

$$d\Big[\langle \phi, e^i \cdot \psi\rangle\, e_i \lrcorner \mu\Big] = \Big(\langle \phi, \mathscr{D}\psi\rangle - \langle \mathscr{D}\phi, \psi\rangle\Big)\mu. \tag{3.37}$$

Proof. Equation (3.36) is just a reformulation of (3.23) using the definition of V^*. Equation (3.37) is the local form of Green's formula (3.7). □

We can now derive the integrated version of the Weitzenböck formula.

Proposition 3.38. *Let $\Omega \subset M$ be a compact domain with smooth boundary. For every $\psi \in \Gamma(S)$,*

$$\int_\Omega \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi\rangle - |\mathscr{D}\psi|^2\Big)\,\mu = -\frac{1}{2}\int_{\partial\Omega} \big\langle \psi,\, [e^i, e^j]\cdot V_j\psi\big\rangle\, e_i \lrcorner \mu. \tag{3.39}$$

Equivalently, as an identity of differential forms on M,

$$\Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi\rangle - |\mathscr{D}\psi|^2\Big)\mu = -\frac{1}{2}\, d\Big[\big\langle \psi,\, [e^i, e^j]\cdot V_j\psi\big\rangle\, e_i \lrcorner \mu\Big]. \tag{3.40}$$

Proof. Take $\phi = \psi$ in (3.36):

$$d\Big[\langle \psi, V_i\psi\rangle\, e_i \lrcorner \mu\Big] = \Big(|V\psi|^2 - \langle \psi, V^*V\psi\rangle\Big)\mu. \tag{3.41}$$

Next take $\phi = \psi$ and ψ replaced by $\mathscr{D}\psi$ in (3.37):

$$d\Big[\langle \psi, e^i \cdot \mathscr{D}\psi\rangle\, e_i \lrcorner \mu\Big] = \Big(\langle \psi, \mathscr{D}^2\psi\rangle - |\mathscr{D}\psi|^2\Big)\mu. \tag{3.42}$$

Adding (3.41) and (3.42), and then using $\mathscr{D}^2 = V^*V + \mathcal{R}$, gives

$$d\Big[\big\langle \psi,\, V_i\psi + e^i \cdot \mathscr{D}\psi\big\rangle\, e_i \lrcorner \mu\Big] = \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi\rangle - |\mathscr{D}\psi|^2\Big)\mu. \tag{3.43}$$

It remains to simplify the boundary expression. Using the definition $\mathscr{D}\psi = e^j \cdot V_j\psi$, we compute

$$V_i\psi + e^i \cdot \mathscr{D}\psi = V_i\psi + e^i \cdot e^j \cdot V_j\psi = (\delta^{ij} + e^i \cdot e^j)\, V_j\psi.$$

Now the tangential Clifford relations imply

$$\delta^{ij} + e^i \cdot e^j = \frac{1}{2}\big(e^i \cdot e^j - e^j \cdot e^i\big) = \frac{1}{2}[e^i, e^j].$$

Therefore

$$V_i\psi + e^i \cdot \mathscr{D}\psi = \frac{1}{2}[e^i, e^j]\cdot V_j\psi.$$

Substituting this into (3.43) gives (3.40), and Stokes' theorem then yields (3.39). □

If ψ satisfies the hypersurface Dirac equation

$$\mathscr{D}\psi = 0,$$

then Proposition 3.38 becomes

$$\int_\Omega \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi\rangle\Big)\,\mu = -\frac{1}{2}\int_{\partial\Omega} \big\langle \psi,\, [e^i, e^j]\cdot V_j\psi\big\rangle\, e_i \lrcorner \mu. \tag{3.44}$$

By Lemma 3.32, the integrand on the left is pointwise nonnegative under the dominant energy condition. This is the positivity statement that ultimately produces the inequality $E_\ell \geq |P_\ell|$.

4. Harmonic spinors and the proof of $E_\ell \geq |P_\ell|$

Throughout this section we continue to assume that (M, g, h) is a complete oriented asymptotically flat initial data set satisfying the dominant energy condition. The integral identity from Proposition 3.38 shows that any harmonic spinor has a nonnegative bulk term, and the remaining problem is to construct harmonic spinors with prescribed asymptotic constants and to identify the resulting boundary contribution.

4.1. **Weighted Sobolev spaces.** Fix $R > 1$ large enough so that each end M_ℓ contains the exterior of the Euclidean ball $B_R = \{x \in \mathbb{R}^3 : |x| < R\}$ under the chosen asymptotic identification from Definition 1.2. For $r > R$ we write

$$M_{\ell,r} := M_\ell \cap \{x \in \mathbb{R}^3 : |x| > r\}.$$

Choose once and for all a smooth function

$$\sigma : M \longrightarrow [1, \infty)$$

with the following properties:

(i) $\sigma \geq 1$ on M;
(ii) $\sigma = r$ on each $M_{\ell,2R}$;
(iii) $\sigma \equiv 1$ on the compact set $M \setminus \bigcup_{\ell=1}^k M_{\ell,R}$.

Thus σ agrees with the Euclidean radius on the ends and is constant on the compact interior.

Definition 4.1. Let $p \geq 2$, let $s = 0$ or 1, and let δ satisfy

$$\frac{1}{2} - \frac{3}{p} \leq \delta \leq 2 - \frac{3}{p}.$$

The weighted Sobolev space $\mathcal{H}_{s,\delta,p}(M; S)$ is the completion of $C_0^\infty(M; S)$ in the norm

$$\|\phi\|_{0,\delta,p} := \|\sigma^\delta \phi\|_{L^p(M)},$$

and

$$\|\phi\|_{1,\delta,p} := \|\sigma^{1+\delta}\nabla\phi\|_{L^p(M)} + \|\sigma^\delta\phi\|_{L^p(M)}.$$

For notational convenience we set

$$\mathcal{H} := \mathcal{H}_{1,-1,2}, \qquad \|\phi\|_{\mathcal{H}} := \|\phi\|_{1,-1,2}.$$

The role of δ is to measure decay at infinity. If $\phi \in \mathcal{H}_{0,\delta,p}$, then $\sigma^\delta\phi \in L^p$, so positive δ forces ϕ to be small on the ends. If $\phi \in \mathcal{H}_{1,\delta,p}$ and $p > 3$, then the weighted L^p control on both ϕ and $\nabla\phi$ implies pointwise decay through the Sobolev embedding on Euclidean annuli.

4.2. **The Green-operator theorem.** The analytic result needed in this section is the following mapping theorem for the hypersurface Dirac operator. Its proof will be given in Section 5.

Theorem 4.2. *For $p \geq 2$ and $0 < \delta < 2 - \frac{3}{p}$, or for $p = 2$ and $\delta = -1$, the operator*

$$\mathscr{D} : \mathcal{H}_{1,\delta,p}(M; S) \longrightarrow \mathcal{H}_{0,\delta+1,p}(M; S)$$

is an isomorphism with bounded inverse. If $\eta \in \mathcal{H}_{0,\delta+1,p}(M; S) \cap C^\infty(M; S)$, then $\mathscr{D}^{-1}\eta \in C^\infty(M; S)$.

The rest of the section uses Theorem 4.2 as input.

4.3. **Harmonic spinors with prescribed asymptotic constants.** Let

$$\{\psi_{\infty,\ell}\}_{\ell=1}^k$$

be constant spinors on the asymptotic ends M_ℓ in the sense of Definition 2.16.

Theorem 4.3. *There exists a unique smooth spinor ψ on M with the following properties:*

(i) $\mathscr{D}\psi = 0$;
(ii) for every $\varepsilon > 0$,

$$\lim_{r\to\infty} r^{1-\varepsilon} \, |\psi - \psi_{\infty,\ell}| = 0 \qquad \text{on each end } M_\ell;$$

(iii)

$$\int_M \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle \Big) \, \mu = 4\pi \sum_{\ell=1}^k \Big(E_\ell \langle \psi_{\infty,\ell}, \psi_{\infty,\ell} \rangle + \big\langle \psi_{\infty,\ell}, (P_\ell)_j \, dx^0 \cdot dx^j \cdot \psi_{\infty,\ell} \big\rangle \Big). \tag{4.4}$$

Here dx^0 and dx^1, dx^2, dx^3 denote the standard coframe on $\mathbb{R}^{3,1}$ used in the asymptotic trivialization of the spinor bundle on each end.

Proof. For each end M_ℓ, choose a smooth function

$$0 \leq f_R^{(\ell)} \leq 1$$

such that

$$f_R^{(\ell)} \equiv 1 \quad \text{on } M_{\ell,3R}, \qquad f_R^{(\ell)} \equiv 0 \quad \text{on } M \setminus M_{\ell,2R}.$$

Set

$$\psi_0 := \sum_{\ell=1}^k f_R^{(\ell)} \psi_{\infty,\ell}. \tag{4.5}$$

By construction, ψ_0 is smooth on M, and on each end it agrees with the constant spinor $\psi_{\infty,\ell}$ outside a compact set.

Because the asymptotic decay assumptions imply that the coefficients of the spin connection V are $O(r^{-2})$, we have

$$\mathscr{D}\psi_0 = O(r^{-2})$$

on each end. Hence

$$\mathscr{D}\psi_0 \in L^2(M; S) \cap \mathcal{H}_{0,\delta+1,p}(M; S) \qquad \text{for all } 2 \leq p < \infty, \quad 0 < \delta < 1 - \frac{3}{p}.$$

Applying Theorem 4.2 in the Hilbert case $(p,\delta) = (2,-1)$ and also in each weighted case with

$$2 \le p < \infty, \qquad 0 < \delta < 1 - \frac{3}{p},$$

we obtain a unique smooth spinor

$$\psi_1 \in \mathcal{H} \cap \mathcal{H}_{1,\delta,p}(M;S)$$

satisfying

$$\mathscr{D}\psi_1 = -\,\mathscr{D}\psi_0. \tag{4.6}$$

Set

$$\psi := \psi_0 + \psi_1. \tag{4.7}$$

Then

$$\mathscr{D}\psi = \mathscr{D}\psi_0 + \mathscr{D}\psi_1 = 0,$$

so ψ is harmonic. This proves (i).

To prove (ii), fix $\varepsilon > 0$. Choose $p > 3$ and δ so that

$$0 < \delta < 1 - \frac{3}{p}, \qquad 1 - \varepsilon < \delta.$$

Since $\psi_1 \in \mathcal{H}_{1,\delta,p}(M;S)$, the weighted quantity $\sigma^\delta \psi_1$ belongs to L^p, and $\sigma^{1+\delta}\nabla\psi_1$ also belongs to L^p. On each dyadic annulus in an end, rescaling reduces this to an ordinary $W^{1,p}$ bound on a fixed Euclidean annulus. Because $p > 3$, the Sobolev embedding $W^{1,p} \hookrightarrow C^0$ shows that

$$r^\delta |\psi_1| \longrightarrow 0 \qquad \text{on each end.}$$

Since $\delta > 1 - \varepsilon$, it follows that

$$r^{1-\varepsilon}|\psi_1| = r^{1-\varepsilon-\delta}\, r^\delta |\psi_1| \longrightarrow 0.$$

Outside a compact set on the end M_ℓ, one has $\psi_0 = \psi_{\infty,\ell}$, hence

$$\psi - \psi_{\infty,\ell} = \psi_1$$

there. Therefore

$$\lim_{r\to\infty} r^{1-\varepsilon}|\psi - \psi_{\infty,\ell}| = 0 \qquad \text{on each end } M_\ell,$$

which proves (ii).

It remains to prove (iii). Since $\mathscr{D}\psi = 0$, Proposition 3.38 gives

$$\int_M \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi\rangle\Big)\,\mu = -\int_M \left[d\eta + \frac{1}{2}\, d\Big(\big\langle \psi_0, [e^i, e^j]\cdot V_j\psi_0\big\rangle\, e_i \lrcorner \mu\Big)\right], \tag{4.8}$$

where η is the 2-form

$$\eta := \frac{1}{2}\Big(\big\langle \psi_1, [e^i, e^j]\cdot V_j\psi\big\rangle + \big\langle \psi_0, [e^i, e^j]\cdot V_j\psi_1\big\rangle\Big) e_i \lrcorner \mu. \tag{4.9}$$

The left-hand side of (4.8) is real, so only the real part of the right-hand side contributes. In the local frame used in Section 3, a direct calculation gives

$$d\Big(\big\langle \psi_0, [e^i, e^j]\cdot \psi_1\big\rangle\, e_i \lrcorner \mu\Big) = \Big(\big\langle V_i\psi_0, [e^i, e^j]\cdot \psi_1\big\rangle + \big\langle \psi_0, [e^i, e^j]\cdot V_j\psi_1\big\rangle\Big) e_i \lrcorner \mu. \tag{4.10}$$

Hence, up to an exact differential whose integral may be ignored, we may replace η in (4.8) by the 2-form

$$\vartheta := \frac{1}{2}\Big(\langle \psi_1, [e^i, e^j]\cdot V_j\psi\rangle - \langle V_i\psi_0, [e^i, e^j]\cdot \psi_1\rangle\Big) e_i \lrcorner \mu. \tag{4.11}$$

We claim that

$$d\vartheta \in L^1(M). \tag{4.12}$$

To verify this, differentiate (4.11) in the frame of Section 3 and use the relations

$$\mathscr{D}\psi = 0, \qquad \mathscr{D}\psi_1 = -\mathscr{D}\psi_0, \qquad V_i = \nabla_i + \frac{1}{2}h_{ij}e^j\cdot e^0\cdot, \qquad V_iV_j - V_jV_i = \Omega_{ij},$$

where Ω_{ij} is the curvature of the spin connection. After expanding the derivative and collecting terms, one obtains a pointwise estimate of the form

$$|d\vartheta| \le C\Big(|V\psi_1|^2 + |V\psi|^2 + |\psi_1|\,|\Omega|\,|\psi| + |\psi_1|\,|\nabla h|\,|\psi| + |\nabla h|\,|\psi_0|\,|\psi_1| + |V\psi_0|^2\Big)\mu. \tag{4.13}$$

Each term on the right-hand side of (4.13) is integrable. Indeed,

$$V\psi_1 \in L^2(M), \qquad V\psi \in L^2(M)$$

because $\psi_1 \in \mathcal{H}$ and $\psi = \psi_0 + \psi_1$ satisfies the identity in Section 3. Also, $V\psi_0 \in L^2(M)$ because $V\psi_0 = O(r^{-2})$ on each end. Further,

$$|\Omega| = O(r^{-3}), \qquad |\nabla h| = O(r^{-3}),$$

while ψ_0 is bounded and ψ_1 satisfies the decay statement from (ii). Therefore the products

$$|\psi_1|\,|\Omega|\,|\psi|, \qquad |\psi_1|\,|\nabla h|\,|\psi|, \qquad |\nabla h|\,|\psi_0|\,|\psi_1|$$

belong to $L^1(M)$. This proves (4.12).

Choose a cutoff function ω_p with the following properties:

(a) $0 \le \omega_p \le 1$;
(b) $\omega_p = 0$ on $\bigcup_{\ell=1}^k M_{\ell,2p}$;
(c) $\omega_p = 1$ on the complement of $\bigcup_{\ell=1}^k M_{\ell,p}$;
(d) $|d\omega_p| \le 2/p$.

Since $d\vartheta \in L^1(M)$, the dominated convergence theorem gives

$$\int_M d\vartheta = \lim_{p\to\infty}\int_M \omega_p\, d\vartheta = \lim_{p\to\infty}\int_M \big(d(\omega_p\vartheta) - d\omega_p\wedge\vartheta\big) = -\lim_{p\to\infty}\int_M d\omega_p\wedge\vartheta.$$

For sufficiently large p, the support of $d\omega_p$ lies in the region where $\omega_p = (1-\omega_p/2)^2$, and $|d\omega_p| \le 2\sigma^{-1}$. Using (4.11), we therefore obtain

$$\begin{aligned}\left|\int_M d\omega_p\wedge\vartheta\right| &\le 2\int_M \sigma^{-1}(1-\omega_p)^{1/2}|\vartheta|\,\mu \\ &\le C\,\|\sigma^{-1}(1-\omega_p)^{1/2}\psi_1\|_{L^2}\Big(\|V\psi\|_{L^2} + \|V\psi_0\|_{L^2}\Big).\end{aligned}$$

Because $\sigma^{-1}\psi_1 \in L^2(M)$ and $1-\omega_p \to 0$ pointwise with $0 \le 1-\omega_p \le 1$, the dominated convergence theorem shows that

$$\|\sigma^{-1}(1-\omega_p)^{1/2}\psi_1\|_{L^2} \longrightarrow 0.$$

Hence

$$\int_M d\vartheta = 0.$$

Since $\eta - \vartheta$ differs by an exact differential, we conclude that

$$\int_M d\eta = 0. \tag{4.14}$$

We are therefore left with

$$\int_M \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi\rangle\Big)\,\mu = -\frac{1}{2}\int_M d\Big(\langle \psi_0, [e^i, e^j]\cdot V_j\psi_0\rangle\, e_i \lrcorner \mu\Big). \tag{4.15}$$

The right-hand side depends only on the asymptotic data. To evaluate it, work on one end M_ℓ in the asymptotic coordinates $x = (x^1, x^2, x^3)$. Under the asymptotic trivialization, the Hermitian structure determined by $e^0\cdot$ corresponds to that determined by $dx^0\cdot$, and the spin connection takes the form

$$V_j\psi_{\infty,\ell} = \partial_j\psi_{\infty,\ell} - \frac{1}{4}\,\Gamma_{kjm}\,dx^k\cdot dx^m\cdot\psi_{\infty,\ell} - \frac{1}{2}\,h_{jk}\,dx^0\cdot dx^k\cdot\psi_{\infty,\ell}. \tag{4.16}$$

Since $\psi_{\infty,\ell}$ is constant,

$$\partial_j\psi_{\infty,\ell} = 0.$$

Also, because the end is asymptotically flat, there is an orthonormal coframe $\{e^i\}$ with

$$|e^i - dx^i| = O(r^{-1}),$$

so the boundary limit is unchanged if e^i is replaced by dx^i.

Substituting (4.16) into the boundary term and using the symmetry $\Gamma_{kjm} = \Gamma_{mjk}$, we find

$$\begin{aligned} &-\frac{1}{2}\int_{S_r}\big\langle \psi_{\infty,\ell}, [dx^i, dx^j]\cdot V_j\psi_{\infty,\ell}\big\rangle\, d\Omega^i \\ &\quad = -\frac{1}{2}\int_{S_r}\Big\langle \psi_{\infty,\ell}, \Big(-\frac{1}{4}\,\Gamma_{kjm}[dx^i, dx^j]\cdot dx^k\cdot dx^m - \frac{1}{2}\,h_{jk}[dx^i, dx^j]\cdot dx^0\cdot dx^k\Big)\psi_{\infty,\ell}\Big\rangle\, d\Omega^i. \end{aligned} \tag{4.17}$$

The Clifford algebra identities

$$\Gamma_{kjm}[dx^i, dx^j]\cdot dx^k\cdot dx^m = 2(\partial_j g_{ij} - \partial_i g_{jj})\,\mathrm{Id}$$

and

$$h_{jk}[dx^i, dx^j]\cdot dx^0\cdot dx^k = 2(h_{ik} - \delta_{ik}h_{jj})\,dx^0\cdot dx^k$$

then give

$$\begin{aligned} -\frac{1}{2}\int_{S_r}\big\langle \psi_{\infty,\ell}, [dx^i, dx^j]\cdot V_j\psi_{\infty,\ell}\big\rangle\, d\Omega^i &= \frac{1}{4}\int_{S_r}(\partial_j g_{ij} - \partial_i g_{jj})\langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle\, d\Omega^i \\ &\quad + \frac{1}{2}\int_{S_r}(h_{ik} - \delta_{ik}h_{jj})\big\langle \psi_{\infty,\ell}, dx^0\cdot dx^k\cdot\psi_{\infty,\ell}\big\rangle\, d\Omega^i. \end{aligned} \tag{4.18}$$

Passing to the limit $r\to\infty$ and using Definition 1.14, we obtain

$$-\frac{1}{2}\lim_{r\to\infty}\int_{S_r}\big\langle \psi_{\infty,\ell}, [dx^i, dx^j]\cdot V_j\psi_{\infty,\ell}\big\rangle\, d\Omega^i = 4\pi\Big(E_\ell\langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle + \big\langle \psi_{\infty,\ell}, (P_\ell)_k\, dx^0\cdot dx^k\cdot\psi_{\infty,\ell}\big\rangle\Big).$$

Summing over the finitely many ends and combining this with (4.15) proves (iii). □

4.4. **A vanishing lemma.** The next lemma is the point at which the decay of the second fundamental form enters in an essential way. The Weitzenböck formula gives $V\theta = 0$ for a decaying harmonic spinor θ, but V is not compatible with the positive definite inner product. One must therefore estimate the growth of $|\theta|$ directly.

Lemma 4.19. *Let θ be a smooth spinor on M satisfying*

$$V\theta = 0.$$

Then the following hold.

(a) If $\theta(x) \to 0$ along some path in one asymptotic end, then $\theta \equiv 0$ on M.
(b) If $\{\theta^\alpha\}$ is a finite collection of smooth V-parallel spinors which is linearly independent on one end, then $\{\theta^\alpha\}$ is linearly independent at every point of M.

Proof. Set

$$u := |\theta| = \langle \theta, \theta \rangle^{1/2}.$$

We first compute $d(u^2)$. Using the definition of the positive definite inner product,

$$u^2 = \langle \theta, \theta \rangle = (e^0 \cdot \theta, \theta).$$

Differentiate in the direction e_i:

$$e_i(u^2) = \big((V_i e^0) \cdot \theta, \theta\big) + \langle V_i\theta, \theta \rangle + \langle \theta, V_i\theta \rangle.$$

Because $V_i\theta = 0$, only the first term remains. By Lemma 3.10,

$$V_i e^0 = -\, h_{ij} e^j,$$

so

$$e_i(u^2) = -\, h_{ij}\, (e^j \cdot \theta, \theta).$$

Therefore

$$d(u^2) = -\, h_{ij}\, (e^j \cdot \theta, \theta)\, e^i.$$

Since Clifford multiplication by a tangential covector has operator norm 1 with respect to $\langle \cdot\,, \cdot \rangle$, we have

$$|(e^j \cdot \theta, \theta)| \le |\theta|^2 = u^2.$$

Hence

$$|d(u^2)| \le |h|\, u^2. \tag{4.20}$$

On the complement of the zero set of θ, divide by u^2 to obtain

$$|d(\log u)| \le \frac{1}{2}|h|.$$

Because $h = O(r^{-2})$, there exists a constant $C > 0$ such that

$$|d(\log u)| \le \frac{C}{r^2}$$

outside a compact set. Let $\gamma : [0, L) \to M$ be any smooth path tending to infinity in an end. Integrating the differential inequality along γ from a point $x_0 = \gamma(0)$ to $x = \gamma(t)$ gives

$$\log u(x) \ge \log u(x_0) - C \int_{\gamma|_{[0,t]}} r^{-2}\, ds.$$

Along a path contained in an asymptotic end, the integral of r^{-2} is bounded by a constant times $1/r(x_0) - 1/r(x)$. Hence

$$u(x) \ge u(x_0) \exp\Big(-C\big(r(x_0)^{-1} - r(x)^{-1}\big)\Big).$$

Now suppose $u(x) \to 0$ along some path to infinity. Letting $x \to \infty$ along that path gives

$$0 \geq u(x_0) e^{-Cr(x_0)^{-1}},$$

so $u(x_0) = 0$. Since x_0 was arbitrary, $\theta \equiv 0$. This proves (a).

For (b), suppose $\sum_\alpha c_\alpha \theta^\alpha$ vanishes at some point $x_0 \in M$. The same linear combination is V-parallel, and because the family $\{\theta^\alpha\}$ is linearly independent on one end, this combination is not identically zero there. In particular, along some path in that end it tends to a nonzero limit. By part (a), a nontrivial V-parallel spinor cannot vanish at any point. This contradiction proves linear independence everywhere. □

4.5. **Proof of the positive energy theorem.** We now prove Theorem 1.20.

Proof of Theorem 1.20. Fix an end M_m. Choose constant spinors on the ends so that

$$\psi_{\infty,\ell} = 0 \quad \text{for } \ell \neq m, \qquad \psi_{\infty,m} \neq 0.$$

Let ψ be the corresponding harmonic spinor given by Theorem 4.3.

Consider the Hermitian endomorphism of the fiber S_∞ over the asymptotic model of M_m,

$$\mathcal{P}_m := (P_m)_j \, dx^0 \cdot dx^j.$$

Because each $dx^0 \cdot dx^j$ is Hermitian with respect to the positive inner product, $\mathcal{P}_m$ is Hermitian. Its square is

$$\mathcal{P}_m^2 = (P_m)_j (P_m)_k \, dx^0 \cdot dx^j \cdot dx^0 \cdot dx^k.$$

Since $dx^0 \cdot dx^j = -dx^j \cdot dx^0$ and $(dx^0 \cdot)^2 = \mathrm{Id}$,

$$dx^0 \cdot dx^j \cdot dx^0 \cdot dx^k = -\, dx^j \cdot dx^k.$$

As in the proof of Lemma 3.32, the off-diagonal terms vanish and the diagonal terms give

$$\mathcal{P}_m^2 = |P_m|^2 \, \mathrm{Id}.$$

Therefore the eigenvalues of $\mathcal{P}_m$ are $\pm |P_m|$.

Choose $\psi_{\infty,m}$ to be an eigenvector of $\mathcal{P}_m$ with eigenvalue $-|P_m|$. Then

$$\langle \psi_{\infty,m}, \mathcal{P}_m \psi_{\infty,m} \rangle = -|P_m| \, \langle \psi_{\infty,m}, \psi_{\infty,m} \rangle.$$

Substituting this choice into (4.4), we obtain

$$\int_M \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle \Big) \, \mu = 4\pi \big(E_m - |P_m| \big) \langle \psi_{\infty,m}, \psi_{\infty,m} \rangle.$$

By Lemma 3.32, the integrand on the left-hand side is pointwise nonnegative. Since $\psi_{\infty,m} \neq 0$, the inner product on the right is strictly positive. Therefore

$$E_m - |P_m| \geq 0.$$

Because the end M_m was arbitrary, this proves

$$E_\ell \geq |P_\ell| \qquad \text{for every end } M_\ell.$$

□

5. Proof of the Green-operator theorem

In this section we prove Theorem 4.2. The argument follows the chain of propositions used by Parker–Taubes: first one proves boundedness and injectivity of

$$\mathscr{D} : \mathcal{H}_{1,\delta,p}(M;S) \longrightarrow \mathcal{H}_{0,\delta+1,p}(M;S),$$

then one obtains a coercive estimate in the Hilbert case $(p,\delta) = (2,-1)$, next solves $\mathscr{D}\phi = \eta$ for compactly supported η, and finally proves a weighted $W^{1,p}$ estimate and passes to general right-hand sides by approximation.

Throughout the section σ is the weight function fixed in Section 4, and for an open set $U \subset M$ we write

$$\|\phi\|_{L^p(U)} := \left(\int_U |\phi|^p \, \mu\right)^{1/p}.$$

When $U = M$ we suppress it from the notation. We also write

$$\|\phi\|_{s,\delta,p;U}$$

for the weighted Sobolev norm obtained by restricting the integrals in Definition 4.1 to U.

5.1. Boundedness of V and $\mathscr{D}$.

We begin with the elementary mapping property of the two first-order operators that appear in the argument.

Proposition 5.1. *Let $p \geq 2$, and assume either*

$$0 < \delta < 2 - \frac{3}{p},$$

or

$$p = 2, \qquad \delta = -1.$$

Then

$$V : \mathcal{H}_{1,\delta,p}(M;S) \longrightarrow \mathcal{H}_{0,\delta+1,p}(M;S)$$

and

$$\mathscr{D} : \mathcal{H}_{1,\delta,p}(M;S) \longrightarrow \mathcal{H}_{0,\delta+1,p}(M;S)$$

are bounded linear maps.

Proof. By Section 2,

$$V_i = \nabla_i + \frac{1}{2}\, h_{ij}\, e^j \cdot e^0 \cdot = \nabla_i - \frac{1}{2}\, h_{ij}\, e^0 \cdot e^j \cdot .$$

Hence in a local orthonormal frame,

$$|V\phi| \leq |\nabla\phi| + C|h|\,|\phi|.$$

Since $h = O(r^{-2})$ on each end and $\sigma = r$ there, we have

$$|h| \leq C\sigma^{-2}$$

globally on M after enlarging the constant C to absorb the compact part. Multiplying by $\sigma^{1+\delta}$ gives

$$\sigma^{1+\delta}|V\phi| \leq \sigma^{1+\delta}|\nabla\phi| + C\sigma^{\delta-1}|\phi|.$$

Because $\sigma \geq 1$,

$$\sigma^{\delta-1} \leq \sigma^{\delta},$$

so

$$\|\sigma^{1+\delta}V\phi\|_{L^p} \leq \|\sigma^{1+\delta}\nabla\phi\|_{L^p} + C\|\sigma^{\delta}\phi\|_{L^p}.$$

This is exactly the boundedness of V.

For $\mathscr{D}$ we use

$$\mathscr{D}\phi = \sum_{i=1}^{3} e^i \cdot V_i\phi,$$

and the Clifford action has bounded operator norm on each fiber. Thus

$$|\mathscr{D}\phi| \le C|V\phi|,$$

and therefore

$$\|\sigma^{1+\delta}\mathscr{D}\phi\|_{L^p} \le C\|\sigma^{1+\delta}V\phi\|_{L^p} \le C\Big(\|\sigma^{1+\delta}\nabla\phi\|_{L^p} + \|\sigma^{\delta}\phi\|_{L^p}\Big).$$

This proves the proposition on the dense subspace $C_0^\infty(M;S)$, hence on the completion $\mathcal{H}_{1,\delta,p}(M;S)$. □

5.2. **Embedding into $\mathcal{H}$ and the L^2 identity.** The next lemma is the bridge between the weighted L^p spaces and the Hilbert space

$$\mathcal{H} = \mathcal{H}_{1,-1,2}.$$

Lemma 5.2. *Let $p \ge 2$, and assume either*

$$\frac{1}{2} - \frac{3}{p} < \delta \le 2 - \frac{3}{p},$$

or

$$p = 2, \qquad \delta = -1.$$

Then there is a continuous embedding

$$\mathcal{H}_{1,\delta,p}(M;S) \hookrightarrow \mathcal{H}.$$

Moreover, for every $\phi \in \mathcal{H}$,

$$\|\mathscr{D}\phi\|_{L^2}^2 = \|V\phi\|_{L^2}^2 + \int_M \langle \phi, \mathcal{R}\phi\rangle\,\mu. \tag{5.3}$$

In particular,

$$\|\mathscr{D}\phi\|_{L^2} \ge \|V\phi\|_{L^2}. \tag{5.4}$$

Proof. We first prove the embedding for $p > 2$. Set

$$q := \frac{2p}{p-2},$$

so that

$$\frac{1}{2} = \frac{1}{p} + \frac{1}{q}.$$

For $\phi \in C_0^\infty(M;S)$, Hölder's inequality gives

$$\|V\phi\|_{L^2} = \|\sigma^{-(1+\delta)} \cdot \sigma^{1+\delta}V\phi\|_{L^2} \le \|\sigma^{-(1+\delta)}\|_{L^q}\,\|\sigma^{1+\delta}V\phi\|_{L^p},$$

and similarly

$$\|\sigma^{-1}\phi\|_{L^2} = \|\sigma^{-(1+\delta)} \cdot \sigma^{\delta}\phi\|_{L^2} \le \|\sigma^{-(1+\delta)}\|_{L^q}\,\|\sigma^{\delta}\phi\|_{L^p}.$$

Thus it suffices to know that $\sigma^{-(1+\delta)} \in L^q(M)$.

On the compact part of M, this is automatic because $\sigma \equiv 1$. On an end, $\sigma = r$, so

$$\int_{M_{\ell,2R}} \sigma^{-q(1+\delta)}\,\mu \lesssim \int_{2R}^{\infty} r^{-q(1+\delta)} r^2\,dr.$$

This converges exactly when

$$-q(1+\delta)+2<-1,$$

that is,

$$q(1+\delta)>3.$$

Substituting $q=\frac{2p}{p-2}$, this becomes

$$1+\delta>\frac{3(p-2)}{2p}=\frac{3}{2}-\frac{3}{p},$$

or equivalently

$$\delta>\frac{1}{2}-\frac{3}{p}.$$

This is our hypothesis. Therefore $\sigma^{-(1+\delta)}\in L^q(M)$, and the embedding follows for $p>2$.

For $p=2$ and $\delta=-1$ the statement is tautological, since the domain is exactly $\mathcal{H}$.

We now prove (5.3). For $\phi\in C_0^\infty(M;S)$, Proposition 3.38 gives

$$\int_M\Big(|V\phi|^2+\langle\phi,\mathcal{R}\phi\rangle-|\mathscr{D}\phi|^2\Big)\,\mu=0,$$

because the boundary term vanishes for compactly supported spinors. This is exactly (5.3) on $C_0^\infty(M;S)$.

Both sides define continuous quadratic functionals on $\mathcal{H}$: the terms $\|V\phi\|_{L^2}$ and $\|\mathscr{D}\phi\|_{L^2}$ are continuous by Proposition 5.1 in the case $(p,\delta)=(2,-1)$, and

$$\left|\int_M\langle\phi,\mathcal{R}\phi\rangle\,\mu\right|\le\|\mathcal{R}\|_{L^\infty}\,\|\phi\|^2_{L^2_{\mathrm{loc}}}+C\|\sigma^{-1}\phi\|^2_{L^2}$$

because $\mathcal{R}$ is smooth on the compact part and $O(r^{-2})$ on each end. Since $C_0^\infty(M;S)$ is dense in $\mathcal{H}$, (5.3) extends to all $\phi\in\mathcal{H}$.

Finally, $\mathcal{R}\ge 0$ by Lemma 3.32, so (5.4) follows immediately. □

5.3. **Injectivity.** The next proposition is the weighted injectivity statement. Its proof is exactly the place where the vanishing lemma from Section 4 is used.

Proposition 5.5. *Let $p\ge 2$, and assume either*

$$0<\delta<2-\frac{3}{p},$$

or

$$p=2,\qquad\delta=-1.$$

Then

$$\mathscr{D}:\mathcal{H}_{1,\delta,p}(M;S)\longrightarrow\mathcal{H}_{0,\delta+1,p}(M;S)$$

is injective.

Proof. Let $\phi\in\mathcal{H}_{1,\delta,p}(M;S)$ satisfy $\mathscr{D}\phi=0$. By the local elliptic regularity theorem recorded in Appendix B, ϕ is smooth.

By Lemma 5.2, $\phi\in\mathcal{H}$. Applying (5.3) to ϕ and using $\mathscr{D}\phi=0$, we obtain

$$0=\|V\phi\|^2_{L^2}+\int_M\langle\phi,\mathcal{R}\phi\rangle\,\mu.$$

Both terms on the right are nonnegative, so

$$V\phi = 0.$$

It remains to show that a V-parallel spinor in $\mathcal{H}_{1,\delta,p}$ must vanish. If $p = 2$ and $\delta = -1$, then $\phi \in \mathcal{H}$ implies

$$\sigma^{-1}\phi \in L^2(M).$$

If $p > 2$ and $0 < \delta < 2 - \frac{3}{p}$, then

$$\sigma^{\delta}\phi \in L^p(M).$$

On an end M_ℓ, using spherical coordinates and $\sigma = r$, this means

$$\int_{2R}^{\infty} \int_{S^2} r^{p\delta} |\phi(r,\omega)|^p \, r^2 \, d\omega \, dr < \infty.$$

By Fubini's theorem, for almost every $\omega \in S^2$ the function

$$r \longmapsto r^{p\delta+2} |\phi(r,\omega)|^p$$

is integrable on $[2R, \infty)$. In particular, along almost every ray $r \mapsto (r, \omega)$, the values $|\phi(r,\omega)|$ become arbitrarily small for large r.

Thus ϕ is V-parallel and tends to 0 along some path in each end. Lemma 4.19(a) then implies $\phi \equiv 0$. □

5.4. **Exterior L^2 estimates.** The next two lemmas are the L^2 estimates from which the coercive inequality in $\mathcal{H}$ is obtained.

For an end M_ℓ, we continue to write $M_{\ell,r}$ for the region $\{x \in M_\ell : r(x) > r\}$.

Lemma 5.6. *For R sufficiently large, the following hold for all $\phi \in \mathcal{H}$:*

(a) on each end M_ℓ,

$$\|\sigma^{-1}\phi\|_{L^2(M_{\ell,2R})} \le 5 \, \|\nabla\phi\|_{L^2(M_{\ell,2R})}; \tag{5.7}$$

(b) on each end M_ℓ,

$$\|\nabla\phi\|^2_{L^2(M_{\ell,2R})} \le 2 \, \|V\phi\|^2_{L^2(M_{\ell,2R})}. \tag{5.8}$$

Proof. By density it is enough to prove both estimates for $\phi \in C_0^\infty(M; S)$.

For part (a), we work in the Euclidean coordinates on the end. Write $d\mu_3$ for Euclidean volume measure on $\mathbb{R}^3$. The classical Hardy inequality in $\mathbb{R}^3$ gives

$$\int_{M_{\ell,2R}} r^{-2} |\phi|^2 \, d\mu_3 \le 4 \int_{M_{\ell,2R}} |d|\phi||^2 \, d\mu_3.$$

Here $|\phi|$ is the pointwise norm from $\langle \cdot, \cdot \rangle$. Since ∇ is compatible with that norm, Kato's inequality gives

$$|d|\phi|| \le |\nabla\phi|.$$

Hence

$$\int_{M_{\ell,2R}} r^{-2} |\phi|^2 \, d\mu_3 \le 4 \int_{M_{\ell,2R}} |\nabla\phi|^2 \, d\mu_3.$$

Because $g_{ij} = \delta_{ij} + O(r^{-1})$, the Riemannian volume measure μ and the Euclidean measure $d\mu_3$ are uniformly equivalent on $M_{\ell,2R}$ once R is large. Also $\sigma = r$ on the end. Thus, after enlarging R if necessary, the previous inequality becomes

$$\|\sigma^{-1}\phi\|_{L^2(M_{\ell,2R})} \le 5 \, \|\nabla\phi\|_{L^2(M_{\ell,2R})},$$

which is (5.7).

For part (b), use the pointwise relation

$$V_i\phi = \nabla_i\phi + \frac{1}{2}h_{ij}e^j \cdot e^0 \cdot \phi.$$

Hence

$$|\nabla\phi| \le |V\phi| + C|h|\,|\phi|,$$

and therefore

$$|\nabla\phi|^2 \le 2|V\phi|^2 + C|h|^2|\phi|^2.$$

Integrating over $M_{\ell,2R}$ yields

$$\|\nabla\phi\|^2_{L^2(M_{\ell,2R})} \le 2\|V\phi\|^2_{L^2(M_{\ell,2R})} + C\||h|\phi\|^2_{L^2(M_{\ell,2R})}.$$

Because $h = O(r^{-2})$, we have $|h| \le C\sigma^{-2}$ on the end. Thus

$$\||h|\phi\|_{L^2(M_{\ell,2R})} \le C\|\sigma^{-2}\phi\|_{L^2(M_{\ell,2R})} \le CR^{-1}\|\sigma^{-1}\phi\|_{L^2(M_{\ell,2R})},$$

since $\sigma \ge 2R$ on $M_{\ell,2R}$. Applying part (a),

$$\||h|\phi\|_{L^2(M_{\ell,2R})} \le CR^{-1}\|\nabla\phi\|_{L^2(M_{\ell,2R})}.$$

Therefore

$$\|\nabla\phi\|^2_{L^2(M_{\ell,2R})} \le 2\|V\phi\|^2_{L^2(M_{\ell,2R})} + CR^{-2}\|\nabla\phi\|^2_{L^2(M_{\ell,2R})}.$$

Choosing R so large that $CR^{-2} \le \frac{1}{2}$, we may absorb the last term into the left-hand side and obtain (5.8). □

5.5. **A coercive estimate in the Hilbert case.** We now derive the basic estimate in $\mathcal{H} = \mathcal{H}_{1,-1,2}$.

Lemma 5.9. *There exists a constant $c > 0$ such that*

$$\|\phi\|_{\mathcal{H}} \le c\,\|\mathscr{D}\phi\|_{L^2} \qquad \text{for all } \phi \in \mathcal{H}. \tag{5.10}$$

Proof. Fix R large enough that Lemma 5.6 holds. Choose a smooth cutoff function $\beta = \beta_R$ such that

$$0 \le \beta \le 1, \qquad \beta \equiv 1 \text{ on each } M_{\ell,3R}, \qquad \beta \equiv 0 \text{ on } M \setminus \bigcup_{\ell=1}^{k} M_{\ell,2R}.$$

Decompose

$$\phi = \phi_{\text{in}} + \phi_{\text{ex}}, \qquad \phi_{\text{in}} := (1-\beta)\phi, \qquad \phi_{\text{ex}} := \beta\phi.$$

We estimate the interior and exterior pieces separately.

Step 1: the interior piece. The spinor ϕ_{in} is supported in the compact region

$$M_{\text{int}} := M \setminus \bigcup_{\ell=1}^{k} M_{\ell,2R}.$$

On $C_0^\infty(M_{\text{int}}; S)$, the bilinear form

$$B(\varphi,\psi) := \int_{M_{\text{int}}} \langle V\varphi, V\psi\rangle\,\mu$$

is positive definite. Indeed, if $B(\varphi,\varphi) = 0$, then $V\varphi = 0$; since φ has compact support, it vanishes identically on each end, and Lemma 4.19(a) implies $\varphi = 0$.

Standard elliptic theory on the bounded domain M_{int} therefore yields a constant $c_2 = c_2(R)$ such that

$$\|\phi_{\text{in}}\|_{H^1(M_{\text{int}})} \le c_2\|V\phi_{\text{in}}\|_{L^2(M_{\text{int}})}.$$

Since $\sigma \equiv 1$ on M_{int}, the H^1-norm and the $\mathcal{H}$-norm are equivalent there, and hence

$$\|\phi_{\mathrm{in}}\|_{\mathcal{H}} \le c_3 \, \|V\phi_{\mathrm{in}}\|_{L^2} \tag{5.11}$$

for some $c_3 = c_3(R)$.

Step 2: the exterior piece. On the support of $d\beta$, we have $\sigma \asymp R$ and $|d\beta| \le c_4\sigma^{-1}$. Using

$$V(\beta\phi) = d\beta \otimes \phi + \beta V\phi,$$

we obtain

$$\|V\phi_{\mathrm{ex}}\|_{L^2} \le \|d\beta \otimes \phi\|_{L^2} + \|\beta V\phi\|_{L^2} \le c_4\|\sigma^{-1}\phi\|_{L^2(\operatorname{supp} d\beta)} + \|V\phi\|_{L^2}.$$

The support of $d\beta$ is contained in the union of the end regions $M_{\ell,2R}$, so Lemma 5.6(a) gives

$$\|\sigma^{-1}\phi\|_{L^2(\operatorname{supp} d\beta)} \le C\|V\phi\|_{L^2}.$$

Consequently,

$$\|V\phi_{\mathrm{ex}}\|_{L^2} \le c_5\|V\phi\|_{L^2}. \tag{5.12}$$

Now Lemma 5.6(a) and (b), applied to ϕ_{ex}, imply

$$\|\phi_{\mathrm{ex}}\|_{\mathcal{H}} = \|\nabla\phi_{\mathrm{ex}}\|_{L^2} + \|\sigma^{-1}\phi_{\mathrm{ex}}\|_{L^2} \le c_6\|V\phi_{\mathrm{ex}}\|_{L^2}.$$

Combining this with (5.12),

$$\|\phi_{\mathrm{ex}}\|_{\mathcal{H}} \le c_7\|V\phi\|_{L^2}. \tag{5.13}$$

Step 3: control of $V\phi_{\mathrm{in}}$. Since

$$V\phi = V\phi_{\mathrm{in}} + V\phi_{\mathrm{ex}},$$

we have

$$\|V\phi\|_{L^2}^2 = \|V\phi_{\mathrm{in}}\|_{L^2}^2 + \|V\phi_{\mathrm{ex}}\|_{L^2}^2 + 2\Re(V\phi_{\mathrm{in}}, V\phi_{\mathrm{ex}})_{L^2}.$$

Using

$$2|(V\phi_{\mathrm{in}}, V\phi_{\mathrm{ex}})_{L^2}| \le \frac{1}{2}\|V\phi_{\mathrm{in}}\|_{L^2}^2 + 2\|V\phi_{\mathrm{ex}}\|_{L^2}^2,$$

we obtain

$$\|V\phi\|_{L^2}^2 \ge \frac{1}{2}\|V\phi_{\mathrm{in}}\|_{L^2}^2 - \|V\phi_{\mathrm{ex}}\|_{L^2}^2.$$

Using (5.12),

$$\|V\phi_{\mathrm{in}}\|_{L^2} \le c_8\|V\phi\|_{L^2}$$

for some $c_8 = c_8(R)$.

Step 4: conclusion. By (5.11), (5.13), and the previous bound,

$$\|\phi\|_{\mathcal{H}} \le \|\phi_{\mathrm{in}}\|_{\mathcal{H}} + \|\phi_{\mathrm{ex}}\|_{\mathcal{H}} \le c_3\|V\phi_{\mathrm{in}}\|_{L^2} + c_7\|V\phi\|_{L^2} \le c_9\|V\phi\|_{L^2}.$$

Finally, (5.4) gives

$$\|\phi\|_{\mathcal{H}} \le c_9\|\mathscr{D}\phi\|_{L^2},$$

which is (5.10). □

5.6. **The Hilbert-space existence theorem.** The coercive estimate leads to an existence theorem in the Hilbert case.

Proposition 5.14. *For each $\eta \in C_0^\infty(M;S)$ there exists a unique $u \in \mathcal{H}$ such that*

$$\mathscr{D}^2 u = \eta. \tag{5.15}$$

Moreover $\mathscr{D}u \in \mathcal{H}$, and

$$\|\mathscr{D}u\|_{\mathcal{H}} \le c\,\|\eta\|_{L^2}, \tag{5.16}$$

where c is the constant from Lemma 5.9.

Proof. Consider the functional

$$\mathcal{J}(u) := \frac{1}{2}\|\mathscr{D}u\|_{L^2}^2 - \Re\int_M \langle u,\eta\rangle\,\mu$$

on the Hilbert space $\mathcal{H}$.

The map $u \mapsto \mathscr{D}u$ is continuous from $\mathcal{H}$ to $L^2(M;S)$ by Proposition 5.1, so $\mathcal{J}$ is C^1. Its derivative in the direction $v \in \mathcal{H}$ is

$$d\mathcal{J}(u)[v] = \Re\int_M \langle \mathscr{D}u, \mathscr{D}v\rangle\,\mu - \Re\int_M \langle v,\eta\rangle\,\mu.$$

Hence the Euler–Lagrange equation of $\mathcal{J}$ is

$$\int_M \langle \mathscr{D}u, \mathscr{D}v\rangle\,\mu = \int_M \langle v,\eta\rangle\,\mu \qquad \text{for all } v \in \mathcal{H},$$

which is the weak form of $\mathscr{D}^2 u = \eta$.

We now show that $\mathcal{J}$ is coercive. By Cauchy–Schwarz,

$$\left|\int_M \langle u,\eta\rangle\,\mu\right| \le \|u\|_{L^2}\,\|\eta\|_{L^2}.$$

On the compact support of η, the weight σ is bounded, so $\|u\|_{L^2} \le C\|u\|_{\mathcal{H}}$. By Lemma 5.9,

$$\|u\|_{\mathcal{H}} \le c\|\mathscr{D}u\|_{L^2},$$

hence

$$\mathcal{J}(u) \ge \frac{1}{2}\|\mathscr{D}u\|_{L^2}^2 - C\|\mathscr{D}u\|_{L^2}\|\eta\|_{L^2} \ge \frac{1}{2c^2}\|u\|_{\mathcal{H}}^2 - C'\|u\|_{\mathcal{H}}\|\eta\|_{L^2}.$$

Thus $\mathcal{J}(u) \to +\infty$ as $\|u\|_{\mathcal{H}} \to \infty$, so $\mathcal{J}$ is coercive. It is also strictly convex because $u \mapsto \|\mathscr{D}u\|_{L^2}^2$ is strictly convex by Lemma 5.9. Therefore $\mathcal{J}$ has a unique minimizer $u \in \mathcal{H}$, and that minimizer satisfies the weak equation (5.15).

By local elliptic regularity, u is smooth. Setting

$$\phi := \mathscr{D}u,$$

we have $\phi \in \mathcal{H}$ by Proposition 5.1 in the Hilbert case, and

$$\mathscr{D}\phi = \mathscr{D}^2 u = \eta.$$

Finally, taking $v = u$ in the weak equation and using Lemma 5.9,

$$\|\mathscr{D}u\|_{L^2}^2 = \int_M \langle u,\eta\rangle\,\mu \le C\|u\|_{\mathcal{H}}\|\eta\|_{L^2} \le Cc\|\mathscr{D}u\|_{L^2}\|\eta\|_{L^2},$$

so

$$\|\mathscr{D}u\|_{L^2} \le Cc\|\eta\|_{L^2}.$$

Applying Lemma 5.9 once more to $\phi = \mathscr{D}u$, and using $\mathscr{D}\phi = \eta$, gives

$$\|\phi\|_{\mathcal{H}} \le c\|\eta\|_{L^2},$$

which is (5.16). □

Proposition 5.17. *For each $\eta \in C_0^\infty(M;S)$ there exists a unique $\phi \in \mathcal{H}$ such that*

$$\mathscr{D}\phi = \eta. \tag{5.18}$$

Moreover,

$$\|\phi\|_{\mathcal{H}} \le c\,\|\eta\|_{L^2}, \tag{5.19}$$

where c is the constant from Lemma 5.9.

Proof. Let $u \in \mathcal{H}$ be the unique solution of $\mathscr{D}^2 u = \eta$ given by Proposition 5.14, and set

$$\phi := \mathscr{D}u.$$

Then $\phi \in \mathcal{H}$ and $\mathscr{D}\phi = \eta$.

If $\widetilde{\phi} \in \mathcal{H}$ is another solution of $\mathscr{D}\widetilde{\phi} = \eta$, then

$$\mathscr{D}(\phi - \widetilde{\phi}) = 0,$$

and Proposition 5.5 in the Hilbert case implies $\phi = \widetilde{\phi}$. The estimate (5.19) is exactly (5.16). □

5.7. **The weighted $W^{1,p}$ estimate.** We now prove the weighted estimate which upgrades the Hilbert-space solution to the full space $\mathcal{H}_{1,\delta,p}$.

Proposition 5.20. *For each $\eta \in C_0^\infty(M;S)$ there exists a unique $\phi \in \mathcal{H}$ such that*

$$\mathscr{D}\phi = \eta.$$

If $p \ge 2$ and $0 < \delta < 2 - \frac{3}{p}$, then

$$\|\phi\|_{1,\delta,p} \le c(\delta,p)\,\|\eta\|_{0,\delta+1,p}. \tag{5.21}$$

Proof. The existence and uniqueness of $\phi \in \mathcal{H}$ follow from Proposition 5.17. Fix such a solution and write

$$\phi = \phi_{\text{in}} + \phi_{\text{ex}}, \qquad \phi_{\text{in}} := (1-\beta)\phi, \qquad \phi_{\text{ex}} := \beta\phi,$$

where $\beta = \beta_R$ is the cutoff fixed in the proof of Lemma 5.9.

We estimate ϕ_{in} and ϕ_{ex} separately.

The interior part. Since ϕ_{in} has support in the bounded domain

$$M_{\text{int}} := M \setminus \bigcup_{\ell=1}^{k} M_{\ell,2R},$$

standard elliptic estimates for first-order systems on bounded domains give

$$\|\nabla\phi_{\text{in}}\|_{L^p(M_{\text{int}})} + \|\phi_{\text{in}}\|_{L^p(M_{\text{int}})} \le c_8\,\|\eta\|_{L^p(M)}. \tag{5.22}$$

Since $\sigma \equiv 1$ on M_{int}, this is equivalent to

$$\|\phi_{\text{in}}\|_{1,\delta,p} \le c_8\,\|\eta\|_{0,\delta+1,p}. \tag{5.23}$$

The exterior part. Fix an asymptotic end $M_\ell \subset \mathbb{R}^3$. As in Section 2, the pullback bundle $\Phi_\ell^* S$ is trivial over the end, so we may regard both ϕ_{ex} and η as spinors on $\mathbb{R}^3$.

Choose $R_0 > 0$ so that $M_{\ell,2R_0} \subset M_\ell$. For $R > R_0$, extend the metric g from the end to a smooth metric g^R on all of $\mathbb{R}^3$ by

$$g^R_{ij}(x) = \delta_{ij} + \beta(3x)\big(g_{ij}(x) - \delta_{ij}\big).$$

Let ∇^R be the spin connection of g^R, let $\{e^i_R\}$ be a g^R-orthonormal coframe, and define

$$\mathscr{D}_R = \sum_{i=1}^{3} e^i_R \cdot \nabla^R_i - \frac{1}{2}\,\beta(3x)\,h_{ij}(x)\,e^0 \cdot e^j_R. \tag{5.24}$$

Then

$$\mathscr{D}_R = \mathscr{D} \qquad \text{on sections supported in } M_{\ell,R}.$$

The asymptotic decay assumptions imply that, given $\varepsilon > 0$, one can choose R sufficiently large so that

$$\sup_{x\in\mathbb{R}^3} |e^i_R - dx^i| < \varepsilon \tag{5.25}$$

and

$$\sup_{x\in\mathbb{R}^3} \sigma(x)^2\big(|F_i(x)| + |h_{ij}(x)|\big)\beta(3x) < \varepsilon, \tag{5.26}$$

where F_i denotes the spin connection coefficients of g on the end. These are precisely the smallness conditions needed to apply the Euclidean end theorem of Cantor used by Parker–Taubes.

Theorem 5.27. *For $p \geq 2$ and $0 < \delta < 2 - \frac{3}{p}$, one can choose R sufficiently large so that*

$$\mathscr{D}_R : \mathcal{H}_{1,\delta,p}(\mathbb{R}^3; S) \longrightarrow \mathcal{H}_{0,\delta+1,p}(\mathbb{R}^3; S)$$

is an isomorphism with bounded inverse.

On $\mathbb{R}^3$, the exterior spinor $\phi_{\text{ex}} = \beta\phi$ satisfies

$$\mathscr{D}_R\phi_{\text{ex}} = d\beta \cdot \phi + \beta\eta. \tag{5.28}$$

Since $\eta \in C^\infty_0(M; S)$, the term $\beta\eta$ belongs to $\mathcal{H}_{0,\delta+1,p}(\mathbb{R}^3; S)$. Also, $d\beta$ is compactly supported, while $\phi \in \mathcal{H} \subset L^p_{\text{loc}}(M; S)$, so

$$d\beta \cdot \phi \in \mathcal{H}_{0,\delta+1,p}(\mathbb{R}^3; S).$$

Thus the right-hand side of (5.28) lies in $\mathcal{H}_{0,\delta+1,p}(\mathbb{R}^3; S)$.

Since $\phi_{\text{ex}} \in \mathcal{H}(\mathbb{R}^3; S)$ and satisfies (5.28), Theorem 5.27 is applicable and gives

$$\|\phi_{\text{ex}}\|_{1,\delta,p;\mathbb{R}^3} \leq c(\delta, p)\,\|d\beta \cdot \phi + \beta\eta\|_{0,\delta+1,p;\mathbb{R}^3}. \tag{5.29}$$

The norms in (5.29) are Euclidean, but because

$$g_{ij} = \delta_{ij} + O(r^{-1}), \qquad \partial_k g_{ij} = O(r^{-2})$$

on the ends, they are equivalent to the weighted norms defined using the metric g.

It remains to estimate the right-hand side of (5.29). The term $\beta\eta$ is bounded directly by $\|\eta\|_{0,\delta+1,p}$. For the cutoff term, since $d\beta$ has compact support and σ is bounded on that support,

$$\|d\beta \cdot \phi\|_{0,\delta+1,p} \leq c_9\,\|\phi\|_{L^p(\operatorname{supp} d\beta)}.$$

Because $\operatorname{supp} d\beta$ is compact and $\phi \in \mathcal{H}$, the Sobolev embedding on bounded domains gives

$$\|\phi\|_{L^p(\operatorname{supp} d\beta)} \leq c_{10}\,\|\phi\|_{\mathcal{H}}.$$

By Proposition 5.17,

$$\|\phi\|_{\mathcal{H}} \le c_{11} \|\eta\|_{L^2}.$$

Since η is compactly supported and $\sigma \equiv 1$ on the compact interior,

$$\|\eta\|_{L^2} \le c_{12} \|\eta\|_{0,\delta+1,p}.$$

Combining these estimates with (5.29), we obtain

$$\|\phi_{\mathrm{ex}}\|_{1,\delta,p} \le c_{13}(\delta,p) \|\eta\|_{0,\delta+1,p}. \tag{5.30}$$

Finally, (5.23) and (5.30) give

$$\|\phi\|_{1,\delta,p} \le \|\phi_{\mathrm{in}}\|_{1,\delta,p} + \|\phi_{\mathrm{ex}}\|_{1,\delta,p} \le c(\delta,p) \|\eta\|_{0,\delta+1,p},$$

which is (5.21). □

5.8. **Surjectivity and completion of the proof.** We now pass from compactly supported smooth right-hand sides to arbitrary right-hand sides in the weighted spaces.

Proposition 5.31. *Let $p \ge 2$, and assume either*

$$0 < \delta < 2 - \frac{3}{p},$$

or

$$p = 2, \qquad \delta = -1.$$

Then

$$\mathscr{D} : \mathcal{H}_{1,\delta,p}(M;S) \longrightarrow \mathcal{H}_{0,\delta+1,p}(M;S)$$

is surjective. If $\eta \in \mathcal{H}_{0,\delta+1,p}(M;S)$ and $\phi \in \mathcal{H}_{1,\delta,p}(M;S)$ satisfies

$$\mathscr{D}\phi = \eta,$$

then the estimate

$$\|\phi\|_{1,\delta,p} \le c(\delta,p) \|\eta\|_{0,\delta+1,p} \tag{5.32}$$

holds.

Proof. First assume $p \ge 2$ and $0 < \delta < 2 - \frac{3}{p}$. Let $\eta \in \mathcal{H}_{0,\delta+1,p}(M;S)$. Choose a sequence $\{\eta_i\} \subset C_0^\infty(M;S)$ converging strongly to η in $\mathcal{H}_{0,\delta+1,p}(M;S)$.

For each i, Proposition 5.20 gives a unique $\phi_i \in \mathcal{H}$ such that

$$\mathscr{D}\phi_i = \eta_i.$$

Since

$$\mathscr{D}(\phi_i - \phi_j) = \eta_i - \eta_j,$$

the estimate (5.21) yields

$$\|\phi_i - \phi_j\|_{1,\delta,p} \le c(\delta,p) \|\eta_i - \eta_j\|_{0,\delta+1,p}.$$

Thus $\{\phi_i\}$ is a Cauchy sequence in $\mathcal{H}_{1,\delta,p}(M;S)$. Let ϕ denote its limit.

By Proposition 5.1,

$$\mathscr{D}\phi_i \longrightarrow \mathscr{D}\phi \qquad \text{in } \mathcal{H}_{0,\delta+1,p}(M;S),$$

so

$$\mathscr{D}\phi = \eta.$$

This proves surjectivity. The estimate (5.32) follows by passing to the limit in (5.21).

For the case $p = 2$ and $\delta = -1$, surjectivity is exactly Proposition 5.17 together with density of $C_0^\infty(M; S)$ in $L^2(M; S) = \mathcal{H}_{0,0,2}(M; S)$. The estimate in this case is Lemma 5.9. □

Proof of Theorem 4.2. Proposition 5.1 shows that

$$\mathscr{D} : \mathcal{H}_{1,\delta,p}(M; S) \longrightarrow \mathcal{H}_{0,\delta+1,p}(M; S)$$

is bounded. Proposition 5.5 gives injectivity, and Proposition 5.31 gives surjectivity together with the a priori estimate for the inverse. Hence $\mathscr{D}$ is an isomorphism with bounded inverse.

If $\eta \in \mathcal{H}_{0,\delta+1,p}(M; S) \cap C^\infty(M; S)$, then $\phi = \mathscr{D}^{-1}\eta$ is smooth by local elliptic regularity. This completes the proof of Theorem 4.2. □